

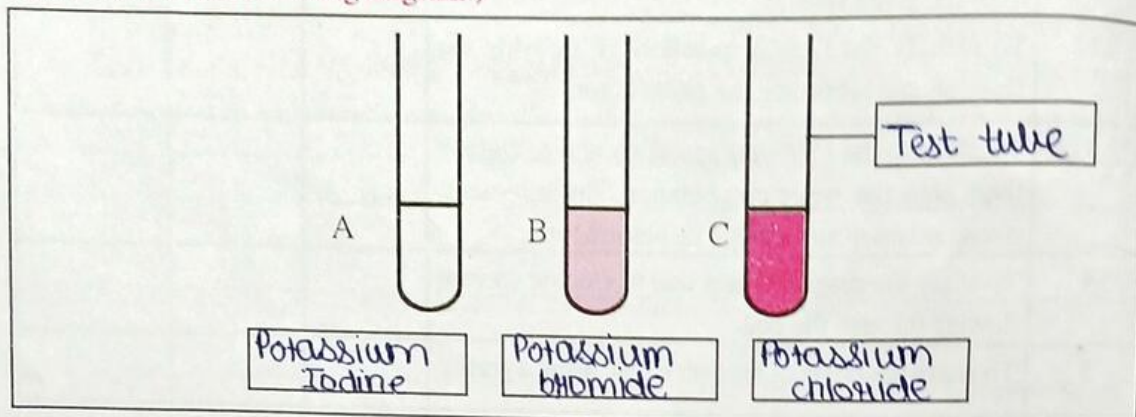
Practical No.1

Aim : To identify the chloride, bromide and iodide ions from the given salts.

Apparatus : Test Tubes, Stand etc.

Chemicals : Silver nitrate, solutions of potassium chloride, potassium bromide and potassium iodide.

Figure : (Label the following diagram.)



Procedure :

1. Take three test tubes and label them as A, B and C.
2. Take about 5ml of solutions of potassium chloride in C, potassium bromide in B and potassium iodide in C.
3. Add about 5ml silver nitrate solution and stir it.
4. Keep test tubes on the stand and observe.

Observations :

Test Tube	Chemical Reaction	Colour of precipitate	ion
A	$KCl + AgNO_3 \rightarrow KNO_3 + AgCl \downarrow$	White	chloride (Cl^-)
B	$KBr + AgNO_3 \rightarrow KNO_3 + AgBr \downarrow$	light yellow	Br.....
C	$KI + AgNO_3 \rightarrow KNO_3 + AgI \downarrow$	bright yellow	

Inference / Conclusion :

1. Ions are precipitated in all the three reaction in the experiment.
2. Elements in the halogen family belonging to 17th group in the periodic table show similarity in their properties.

The halogens are all element that are found in group 17th the periodic table the halogens include this fluorine chlorine bromine iodine and astatine. All of these elements are considered to be reactive non-metals atoms of an element react with other atoms. Based on the 2 number of electrons found in their outer orbit.

Multiple Choice Questions

- Valency of the elements in the halogen group is one
a. one b. two c. three d. four
- The most reactive element in the halogen group is Fluorine ...
a. Astatine b. Iodine c. Chlorine d. Fluorine
- The halogen which is liquid at room temperature is Bromine ...
a. Fluorine b. Astatine c. Bromine d. Iodine
- The metallic character of elements increases in a group from top to bottom.
a. increases b. decreases c. remains constant d. shows indefinite behaviour.
- Valency of elements increases in the beginning and then decreases a period from left to right.
a. Increases b. decreases
c. remains constant d. increases in the beginning and then decreases.

: Exercise :

- Observe the Modern Periodic Table and explain the gradation in reactivity of Halogen family.
In the Modern Periodic Table, elements of group no. 17 are termed as halogens. These elements are very much reactive. They are diatomic in nature (X_2). They are believed to be highly reactive with alkali metals and other metals. The reason behind this is they possess high electro-negative. They can easily accept electrons within the group. Fluorine is considered to be highly reactive while going downwards the group electro-negativity of elements decreases, hence reactivity also decreases.
- What are the similarities in properties of elements in Halogen family?
Similarities in the properties of elements in Halogen family are as follows -
 - They all form acids when combined with hydrogen.
 - They are fairly toxic.
 - They are highly reactive and electronegative.
 - They have seven electrons in their outer shell.
 - They all exist as diatomic molecules, when in their pure form.
- Why does Inert gases placed in Zero group?
Inert gases have their completely filled valence shell with a set of 8 electrons so they have zero valency and thus non-reactive in nature. Hence, inert gases placed in 18th group i.e. in zero group.

Remark and Signature



Practical No. 2

Aim : To identify the type of reaction by studying the reaction and recording the observations.

1. Combustion of magnesium in air.
2. Action of dilute sulphuric acid on zinc.
3. To heat lead nitrate.

Apparatus : Beaker, test tubes, pair of tongs, burner, glass rod, etc.

Chemicals : Zinc dust, magnesium strip, lead nitrate powder, dilute sulphuric acid.

Procedure :

Part I : Combustion of magnesium in air.

1. Hold a piece of magnesium strip on the flame of a burner.
2. Record the observation and write the reaction.

Reaction 1 :



Observations :

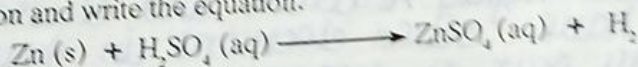
1. The magnesium strip burns with luminous flame.
2. A white coloured powder remains behind.

Reaction	Number of reactants	Number of products	Interesting feature of the reaction	Type of the reaction
$2\text{Mg} + \text{O}_2 \longrightarrow 2\text{MgO}$	2	1	In this reaction when combination magnesium strip burns in the presence of O_2 white coloured powder remains behind.	Reaction...

Part II : Action of dilute sulphuric acid on zinc.

1. Take some zinc dust in a test tube.
2. Add 3 to 4 ml dilute sulphuric acid ($\text{dil H}_2\text{SO}_4$) in it.
3. Record the observation and write the equation.

Reaction 2 :



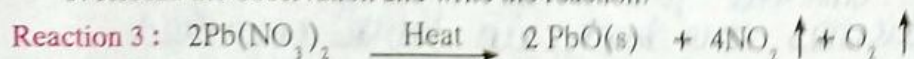
Observations :

1. A colourless gas is liberated.
2. A glowing splinter get extinguished and the gas burns with blue flame production a noise.
3. The zinc powder disappears and a colourless solution is obtained.

Reaction	Number of reactants	Number of products	Interesting feature of the reaction	Type of the reaction
$\text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{ZnSO}_4 + \text{H}_2$	2	2	The gas burns with blue flame and a noise is produced	Displacement Reaction

Part III : To heat lead nitrate.

1. Take a one teaspoonful white coloured lead nitrate powder in a test tube.
2. Heat it on a burner.
3. Hold a moist blue litmus paper at the mouth of the test tube.
4. Hold a glowing splinter at the mouth of the test tube.
5. Record the observation and write the reaction.



Observations :

1. A reddish brown coloured gas turning moist blue litmus paper red is liberated.
2. The glowing splinter keeps on glowing.

Reaction	Number of reactants	Number of products	Interesting feature of the reaction	Type of the reaction
$2\text{Pb}(\text{NO}_3)_2 \xrightarrow{\text{Heat}} 2\text{PbO(s)} + 4\text{NO}_2 \uparrow + \text{O}_2 \uparrow$	1.....	3....	Simple substances are formed.	Decomposition reaction / endothermic reaction.

Inference / Conclusion :

1. Reactions are called 'displacement reactions', when atom or group of atoms in one substance takes place of atom or group of atoms in the other substance to form new substances.
For example, action of dilute sulphuric acid on zinc.
2. Reaction are called 'combination reactions', when a single product is formed from two or more reactants in a chemical reaction.
For example, the reaction between magnesium and oxygen.
3. When two or more simpler substances are formed from a single compound, the reaction is called 'decomposition reaction'. For example, heating lead nitrate.

Multiple Choice Questions

1. Rusting of an iron nail is a combination reaction.
a. combination b. displacement c. decomposition d. double displacement
2. The following change is observed on dipping a litmus paper in aqueous solution of MgO.
☒ a. Red litmus turns blue, therefore MgO is alkaline.
b. Blue litmus turns red, therefore MgO is acidic.
c. No colour change in litmus paper, therefore MgO is neutral.
d. Litmus paper is decolourised, therefore MgO acts as bleaching agent.
3. What is the colour of a solution formed on dipping a piece of zinc in dilute sulphuric acid?
☒ a. colourless b. colourless solution turns red c. black d. red

: Exercise :

1. Why does some liquid medicines kept in dark coloured bottles?

Some liquid medicine are kept in dark coloured bottles because, some medicine contain chemicals which can react with light and start to decompose in the presence of light. If these liquid medicines are kept in dark coloured bottles then they will prevent the medicines to come in contact with light.

2. Why does edible oils shows rancidity when stored for long period?

When edible oils are stored for long period of time then oxidation occurs. There is carbon carbon double bond in fats and oils which is broken down by the oxygen in the atmosphere. As a result, after a long period of time fats and oil become rancid.



In above reaction, write for each reactant that undergoes oxidation or reduction and identify the type of reaction.

In the above reaction BaSO_4 is a reactant which undergoes reduction. Whereas reactant carbon (C) undergoes oxidation.

In this reaction, oxidation and reduction reaction take place simultaneously. Therefore, this is redox reaction.

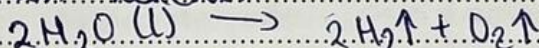
4. Give one example each of displacement, decomposition and combination reaction with explanation.

1] Displacement Reaction -



Here, in this reaction Zn is more reactive than Cu and displaces Cu which is less reactive.

2] Decomposition Reaction -



In this reaction, water decomposes into hydrogen and oxygen gases on passing electric current through it.

3] Combination Reaction - $2\text{Mg} + \text{O}_2 \longrightarrow 2\text{MgO}$

In this reaction 2MgO is formed by combination of Mg and O.

Remark and Signature



Practical No. 3

Aim : To observe the following reactions and to classify them into the types combination, displacement, decomposition and double displacement.

Reaction :

1. Reaction of water with (Calcium Oxide) lime.
2. Effect of heat on ferrous sulphate.
3. Reaction of copper sulphate solution with iron nail.
4. Reaction of solutions of sodium sulphate and barium chloride with each other.

Apparatus : 250 ml beaker, china dish, asbestos sheet, dropper, hard glass test tube, test tube holder, test tube stand, sand paper, burner / spirit lamp, filter paper etc.

Chemicals : Calcium oxide, water, crystals of ferrous sulphate, iron nail / wire scrubber, solution of copper sulphate.

Procedure :

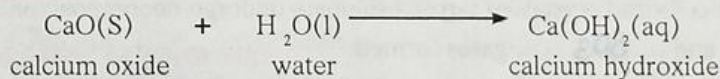
A. Reaction of water with slaked lime.

1. Take about 10 gm lime in a clean and dry china dish, place this dish on an asbestos sheet.
2. Take a little water in a beaker, using dropper sprinkle some water on the lime in the dish.
Record your observations.

Observation :

Sr.No.	Experimental procedure	Observations
1	Note the heat absorbed or evolved during the reaction by touching the dish.	During this reaction heat is evolved.
2	Note whether a gas or vapour is given away in the reaction.	No
3	Note the noise, if, any, produced during the reaction.	No
4	Note the change in the physical state	Solid CaO and liquid H ₂ O get reacted and forms Ca(OH) ₂ an aqueous solution.

Reaction :



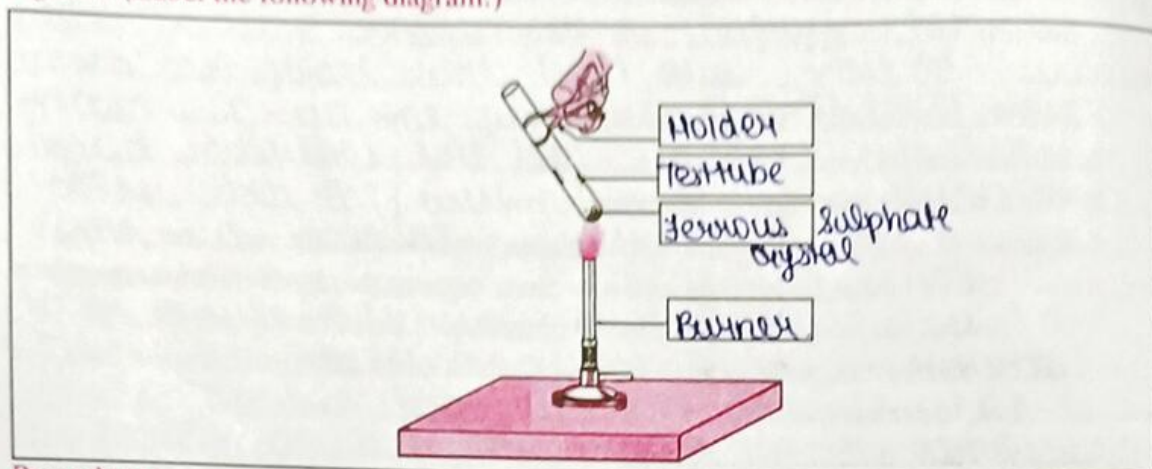
Inference / Conclusion :

The reaction of water with lime is combination reaction. Here calcium oxide and water react to form aqueous solution of calcium hydroxide Ca(OH)₂. Heat is evolved during this reaction, hence it is exothermic reaction.

Procedure :

B. Effect of heat on crystals of ferrous sulphate

Figure : (Label the following diagram.)



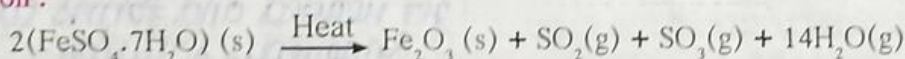
Procedure :

1. Take about 10 gm of powdered crystals of ferrous sulphate in a test tube.
2. Clamp the test tube to a stand and heat it with burner / spirit lamp for 10 minutes.
3. Note the colour of the gas evolved. (Do not smell the gas).
4. Continue heating until the colour of the substance in the test tube changes.
5. Keep the hot test tube on an asbestos sheet. Observe the colour of the substance in it after cooling.

Observation :

Sr.No.	Experiment procedure	Observations
1	Note the original colour of ferrous sulphate.	light green....
2	Note the colour of the gas evolved on heating.	colourless....
3	Observe the colour of the substance in the cold test tube and note it.	Dark Brown...

Reaction :



Inference / Conclusion :

1. On heating the pale green coloured crystals of ferrous sulphate undergo decomposition.
A mixture of SO_2 and SO_3 gases formed.

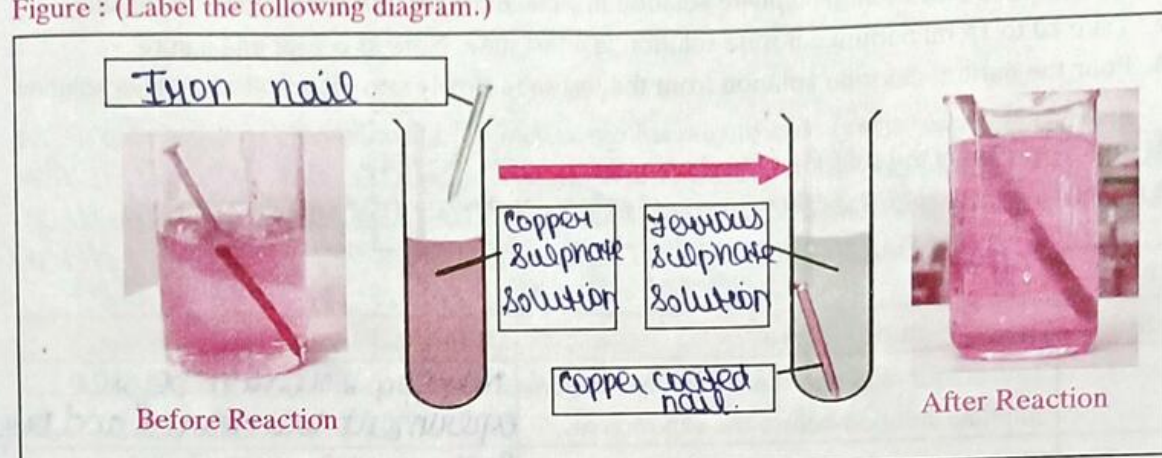
2. A residue of dark brown colour remains in the test tube.

When the ferrous sulphate crystal heated they loses water molecules due to break down of crystal structure to form colourless powder. This is the example of decomposition reaction.

Procedure :

C. Reaction of solution of copper sulphate with iron nail.

Figure : (Label the following diagram.)



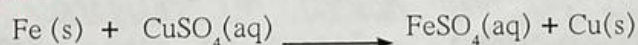
Procedure :

1. Take about 100 ml solution of copper sulphate in a beaker. Note its colour.
2. Take two-three unruined iron nails. Clean them by rubbing with sand paper and wash with water. Note the colour of the nails.
3. Keep the nails immersed in the copper sulphate solution for about fifteen minutes.
4. Observed the change in the colour of the nails and the solution.
5. Remove the nails from the solution after fifteen minutes. Wash them and keep them on a filter paper. Note the changed colour of the nails and the solution.

Observation Table:

Sr.No.	Experiment procedure	Observations
1	Colour of CuSO_4 soln. before the experiment.	Dark Blue
2	Colour of iron nail before the experiment.	Steel Grey
3	Colour of CuSO_4 soln. after the experiment.	Pale Green
4	Colour of iron nail after the experiment.	Reddish Brown

Reaction :



Inference / Conclusion :

1. On immersing the brown coloured iron nails in blue coloured copper sulphate solution, they displace copper... from the copper sulphate solution and their colour becomes Reddish Brown.
2. This is a displacement reaction.

In this reaction, the reddish brown coating on the iron nail shows that copper is deposited on the iron nail that means higher reactive metal displaces less reactive metal.

Procedure :

D. The reaction between sodium sulphate and barium chloride :

1. Take about 20 ml sodium sulphate solution in a clean beaker. Note the colour and nature.
2. Take 10 to 15 ml barium chloride solution in a test tube. Note its colour and nature.
3. Pour the barium chloride solution from the test tube slowly into the sodium sulphate solution in the beaker.
4. Keep on stirring the solution in beaker.
5. Observe the changes occur in beaker and note it down.

Observation Table :

Sr.No.	Experiment procedure	Observations
1	The colour and the nature of the sodium sulphate solution before the experiment.	Na_2SO_4 solution before experiment was clear and transparent.
2	The colour and the nature of the barium chloride solution before the experiment.	BaCl_2 solution before reaction is clear and transparent.
3	The colour and the nature of the mixture resulting on mixing the two solution into each other.	The colour and nature of the mixture is white precipitate is formed.

Reaction :



In this chemical reaction two new compounds are formed by mutual exchange of the components (ions or radicals) of the two compounds. Such reactions are called 'double displacement' reactions.

Inference / Conclusion :

1. In this reaction white coloured insoluble BaSO_4 is formed. As a result of a white coloured precipitate is formed in the beaker.

Multiple Choice Questions

1. The reaction of water with slaked lime is studied by sprinkling a little water into slaked lime.
 - a. putting slaked lime into water taken in a test tube.
 - b. putting a lot of water into slaked lime.
 - c. sprinkling a little water on slaked lime.
 - d. None of the above method.
2. When ferrous sulphate crystals are heated, the residue obtained in the test tube is Red coloured.
 - a. red coloured
 - b. blue coloured
 - c. green coloured
 - d. colourless
3. When sodium sulphate solution reacts with barium chloride solution, the solution after the reaction contains mainly sodium chloride.
 - a. barium sulphate
 - b. sodium chloride
 - c. a and b both
 - d. none of a and b
4. The reaction of iron nail with copper sulphate solution is displacement reaction.
 - a. combination
 - b. decomposition
 - c. displacement
 - d. double displacement

5. The reddish brown colour obtained on the iron nail placed in copper sulphate solution is of Cu.....

a. Cu_2O

b. CuO

c. Cu

d. CuS

: Exercise :

1. "To pour water on calcium oxide." Which is another way to classify this reaction? Explain.
When water is added to calcium oxide (quick lime) it will be converted into calcium hydroxide (slaked lime) with the liberation of enormous amount of heat.
$$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{heat}$$

It is an exothermic reaction.
Calcium hydroxide is an alkali.

2. In experiment (C), speed of reaction increases if we use iron wire scrubber instead of iron nail.

Which factor do you observe that affects the rate of reaction? Explain.

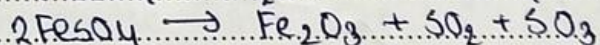
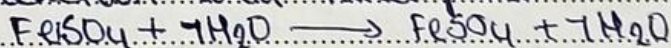
The surface of iron wire scrubber is very high compared to the surface area of iron nail of same mass.

Rate of reaction and area of surface.

$\therefore$ Rate of reaction is affected by surface area.
This reaction, more surface area means more site for reaction to take place. This increases in area increases the speed of the reaction. The factor responsible is surface area.

3. Name the substance which remains in a test tube after heating ferrous sulphate.

Heating ferrous sulphate in a test tube following reaction will take place.



Both SO_2 that is sulphur dioxide and SO_3 that is sulphur trioxide are gases evolved out. Only Fe_2O_3 remained in test tube.

4. Observe double displacement reactions given in the text book and write down the similarities.

The reaction in which two different atoms or group of atoms are mutually exchanged is called double displacement reaction to form precipitate reaction.

Similarities - (i) ions are exchanged
(ii) Precipitate is formed

Remark and Signature

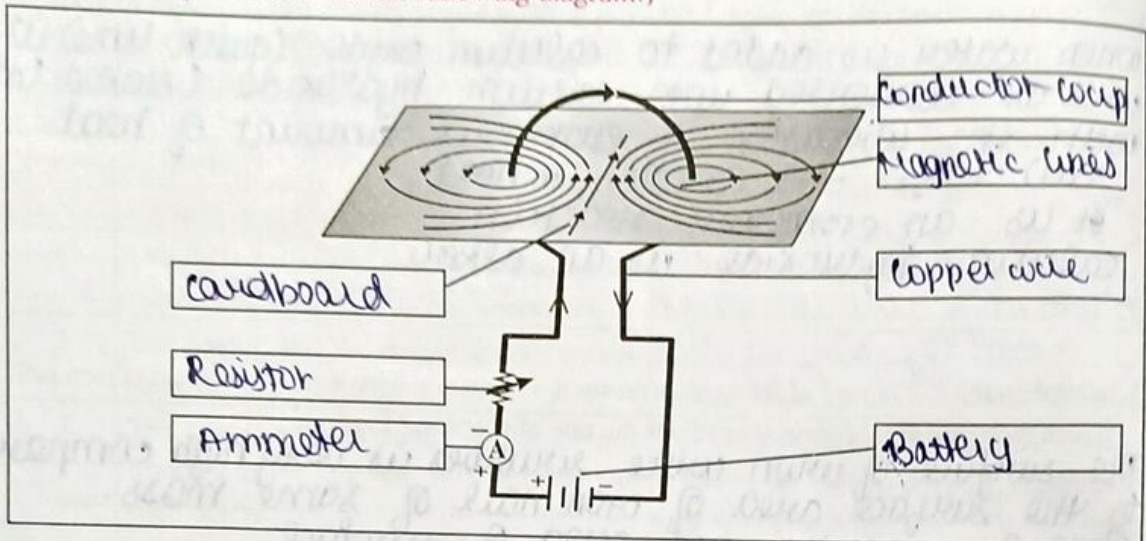


Practical No. 4

Aim : To study the magnetic field due to electric current flowing through the coil.

Apparatus : Insulated copper wire (24 gauge), connecting wires, battery (12V), cardboard/ plastic sheet, iron filings etc.

Figure : (Complete and label the following diagram.)



Procedure :

1. Make a coil around the hollow tube with around 20 to 25 rounds of copper wire.
2. Arrange coil and cardboard / plastic sheet as shown in the figure.
3. Spread some iron filings on the cardboard.
4. Connect both ends of coil to the battery and start the electric current.
5. Keep flicking gently on the cardboard and observed the arrangement of iron filings.
6. Increase the current in circuit gradually and observe the change in the arrangement of iron filings.

Observation :

1. When electric current starts flowing through the coil, magnetic lines are produced at each point on the coil.
2. As we go away from the wire, the concentric circles representing the magnetic lines of forces will become bigger and bigger.
3. As intensity of electric current increases the magnetic lines of force becomes more clearer.

Inference / Conclusion :

1. If the current flows through the coil, magnetic lines of force are produced at each point on the coil.
 2. In above experiment, the intensity of magnetic field at any point by a current flowing through a coil, is dependent on the current and number of turns of the wire.
- At the given point on the cardboard the magnetic field produced by current flowing through the coil is directly proportional to the current and the number of wire forming the coil.

Multiple Choice Questions

- In above experiment ~~magnetic needle~~ would be use to find direction of magnetic lines of force.
 a. bar magnet b. horse shoe magnet
 c. disc magnet ✓ d. magnetic needle
- When electric current is passed through the solenoid, its shows magnetic lines of force similar to a ~~bar magnet~~
 ✓ a. bar magnet. b. horse shoe magnet.
 c. disk magnet. d. spherical magnet.
- In ~~computer~~ appliance Fleming Left Hand rule is not used.
 a. electric fan b. mixer c. computer d. electric generator
- Heating effect of electric current does not observed in ~~electric motor~~
 a. electric oven. b. electric iron. c. electric motor. d. fuse.
- ~~Ohm meter~~ is used to measure electric resistance in a circuit.
 a. Voltmeter b. Galvanometer c. Ammeter d. Ohm meter

: Exercise :

- What is the reason behind covering wires carrying electricity with rubber in some places?

The reason for the covering of wire with rubber is that as rubber is insulator it doesn't conduct electricity so as it would not be covered with insulator like rubber then anyone can get shock and he would supply current which led to his death also.

- What procedure will you follow to study magnetic field, if you are provided with paraffin oil, iron nail, bar magnet, battery, coil conductor wire?

To study the magnetic field, an electromagnet can made using iron nail, wire and battery. By electromagnet magnetic field can be studied by compass magnetism created using two types of magnet - Permanent magnet and temporary magnet - electromagnet.

- Which magnet will you choose among permanent magnet and electromagnet for Industrial purpose? Justify your answer.

The electromagnet could have good uses for industrial purpose, because of its characteristic as a temporary magnet which loses its magnetic field when disconnect from the electric source. Such uses may include the cranes, which are used in dragging metals scrap from a place to another.



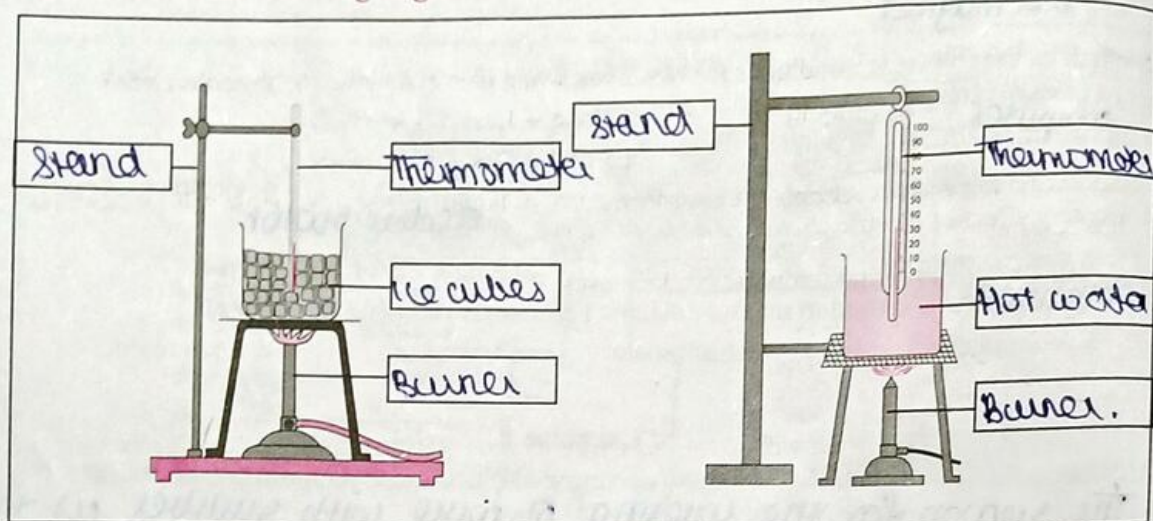
Remark and Signature

Practical No. 5

Aim : To study the effect of heat on ice by using a graph.

Apparatus : Glass beaker, ice, thermometer, stand, tripod stand, spirit lamp or burner.

Figure : (Label the following diagram.)



Procedure :

1. Take a few pieces of ice in a glass beaker.
2. Keep beaker on tripod stand and put it on the stand.
3. Attach a thermometer on the stand in such a way that the thermometer bulb will completely dip in the ice.
4. Measure the temperature of ice.
5. Arrange the apparatus as shown in figure.
6. Heat the ice using a burner or spirit lamp.
7. Record the temperature using the thermometer after every minute.
8. Continue the heating and recording temperature even after ice starts melting and boiling.
9. Draw a graph of temperature versus time.

Observation :

1. Least count of thermometer = ... 1 °C

Observation table :

Time	Temp °C	Time	Temp °C
0	0	5	42
1	0	6	56
2	10	7	72
3	21	8	81
4	30	9	93

Inference / Conclusion :

- Heat energy is absorbed during the transformation of ice into water and water into vapours.
- No rise in temperature during transformation of state.
- The amount of heat energy absorbed at constant temperature by unit mass of a solid to convert into liquid phase is called specific latent heat of fusion.

Multiple Choice Questions

- Which of the following conditions are observed when heat is supplied to the substance?
a. Solid + Liquid, Liquid
b. Solid + Liquid, Liquid, Gas
c. Solid + Gas
d. Solid + Gas, Liquid, Gas
- 'Latent heat is evolved during the transformation of liquids into solids.' How will you explain this statement?
a. Heat released and temperature decreases.
b. Heat absorbed and temperature increases.
c. Heat does not absorbed nor released.
d. Not sure.
- In above experiment, what will be the maximum rise in temperature of water after melting of ice?
a. 27°C b. 0°C c. 4°C d. 100°C
- The increase in temperature of water above its boiling point in the pressure cooker is due to change in
a. pressure b. volume c. mass d. all of these
- The temperature of ice can be decreased below 0°C by mixing in it.
a. saw dust b. sand c. salt d. coal

: Exercise :

- What could be explained from the straight horizontal line observed at the beginning and end of the graph?

The straight horizontal line in the beginning and end of the graph shows that there is no increase or decrease in temperature. Only change in state takes place.

- How could latent heat be explained from the graph? Explain in detail.

In the graph line represents the constant temperature even after giving it heat. This heat is used for weakening the bonds of solid. This temperature remains constant till the total ice change into water. Then temperature starts rising.



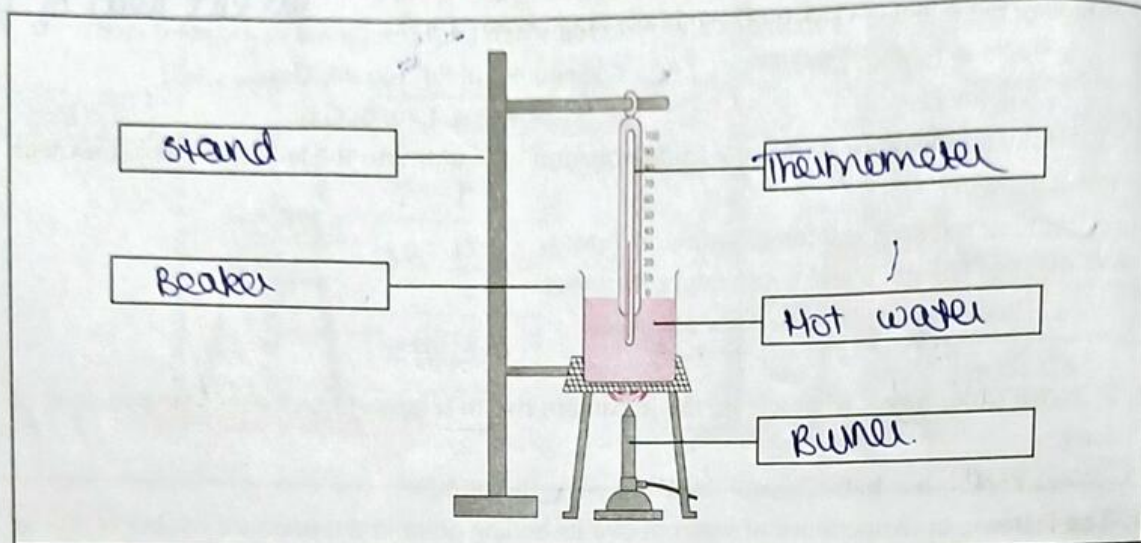
Remark and Signature

Practical No. 6

Aim : To measure temperature of hot water during natural cooling and plot the graph of temperature versus time.

Apparatus : A 250 ml glass beaker, thermometer with gradation from 0°C to 100°C , a clock, etc.

Figure : (Label the following diagram.)



Procedure :

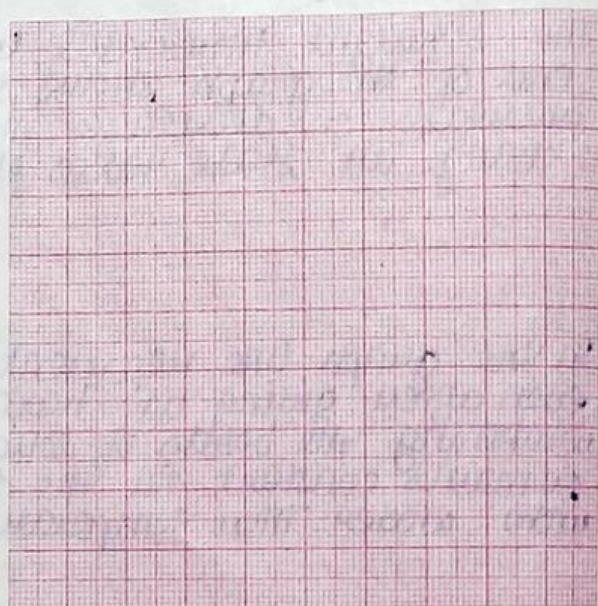
1. Heat the water in the beaker up to 100°C .
2. Put off the burner.
3. When the water reaches 70°C , start measuring the temperature after every minute.
4. Measure the temperature till the water reaches ambient temperature.
5. Plot the graph of temperature versus time (in mins.).

Observation :

1. Initial temperature of water before heating = $^{\circ}\text{C}$

2. Observation Table

Time (mins)	Temperature ($^{\circ}\text{C}$)
0	70°
1	64°
2	58°
3	54°
4	51°
5	43°



Conclusion / Inference :

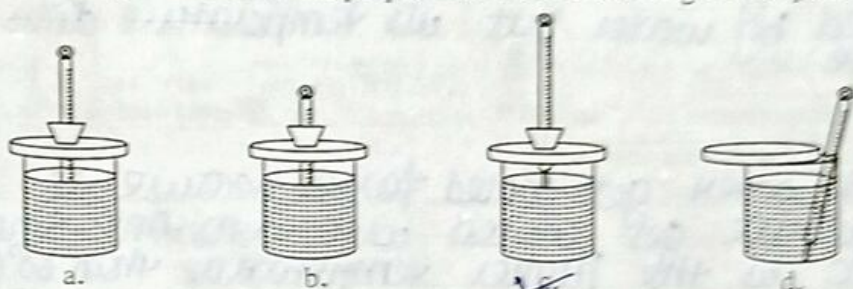
1. The rate of cooling of water is slow when the difference in the temperature of water and the ambience is large.
2. This rate evaporation increases as the temperature of water reduces due to cooling.

Precautions :

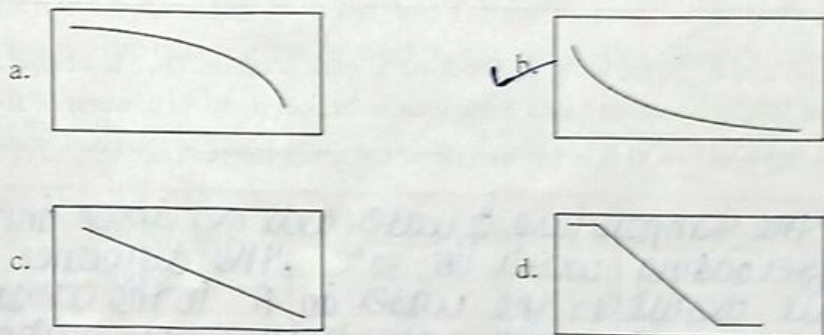
1. Switch off the ceiling fans in the laboratory before the observations.
2. Let the clock run continuously while cooling, do not stop it.

Multiple Choice Questions

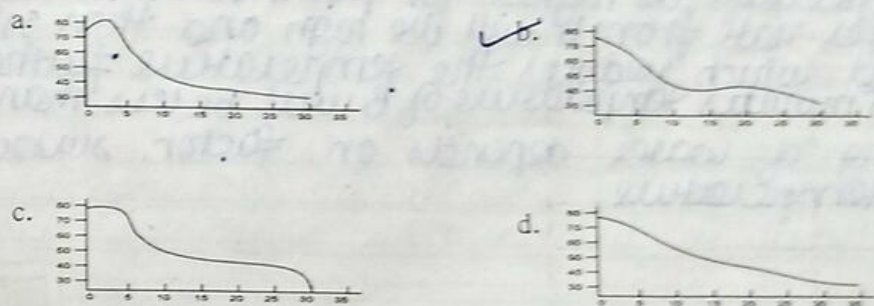
1. Out of the following, is the proper method for measuring the temperature of water.



2. While cooling hot water in natural ambience, it follows type of behaviour out of the following.



3. Following are the temperature - time graphs obtained by 4 students in this experiment. Which one is likely to be correct?



4. Hot liquids gets cooled faster
- a. inside house in winter.
 - b. in the open in winter.
 - c. in the open in summer.
 - d. inside house in summer.

5. Heat is liberated when

- a. water converts into vapour.
- c. water heats.

b. ice melts.

d. water vapours condenses.

: Exercise :

1. What is indicated by time - temperature graph in above activity?

The temperature of water starts raising. It increases upto 100°C . Rise of temperature of water 0°C to 100°C Therefore its temperature does not rise and converts the liquid into gaseous state. Thus heat energy is absorbed by water but its temperature does not change.

2. If water of temperatures 80°C and 60°C is cooled in two different vessels, which water gets cooled faster? Justify your answer.

60°C of water gets cooled faster because all temperature get cooled with constant time and 80°C is the higher temperature than 60°C .

3. In two beakers (A and B) 50 ml water of about 70°C each is taken. Then in beaker A, 50 ml of water of about 30°C is added and after 5 minutes, in beaker B, 50 ml of water of about 30°C is added. What would observe if the temperatures of both beakers recorded immediately? Why?

In A since the temperature of water will get down immediately after adding water at 30°C , the difference in temperature between the water in A to the atmosphere is less so heat transfer rate between the beaker A and atmosphere will reduce whereas in case of beaker B the temperature is higher for five minutes so the heat transfer rate from B will be high and then 30°C water is added which reduces the temperature further. So after 5 minutes temperature of B will be less than A.

4. On which factors does cooling of water depends?

Cooling of water depends on factor surrounding air temperature.

Remark and Signature

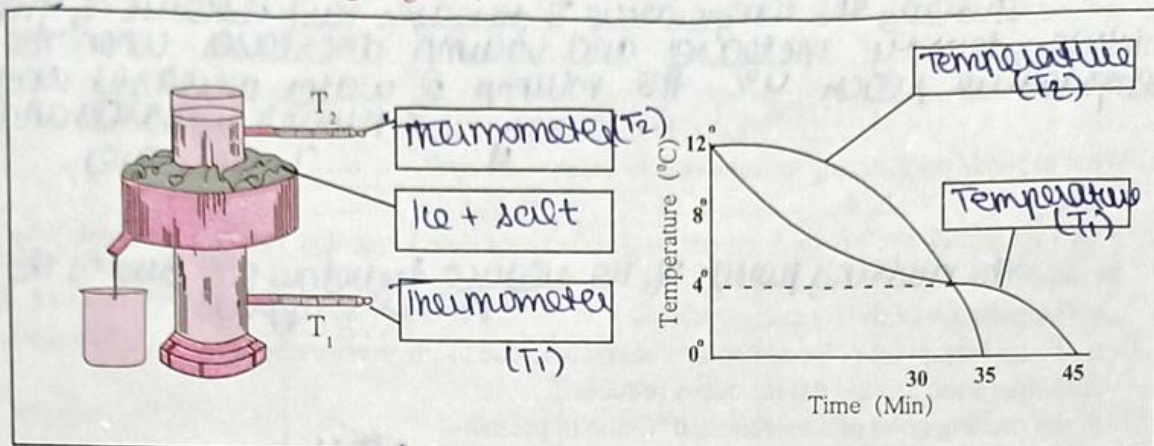


Practical No. 7

Aim : To study anomalous behaviour of water using Hope's apparatus.

Apparatus : Hope's apparatus, ice, common salt, water, thermometer, beaker, etc.

Figure : (Label the following diagram.)



Procedure :

1. Take cold water ($10-12^{\circ}\text{C}$ temp) in the inner cylinder of the Hope's apparatus.
2. In outer wider container, fill in the freezing mixture of ice and salt.
3. Fix the two thermometers T_1 (lower) and T_2 (upper), as shown in the figure.
4. Observe the two temperatures at every minute and make an entry in the observation table below.
5. Study the given graphs and indicate the curves corresponding to T_1 or T_2 .

Observation Table :

No	Time (Minutes)	Temperature (T_1 $^{\circ}\text{C}$)	Temperature (T_2 $^{\circ}\text{C}$)
1	0	10°C	12°C
2	1	9°C	11°C
3	2	8°C	10°C
4	3	7°C	9°C
5	4	6°C	8°C
6	5	5°C	7°C
7	6	4°C	6°C
8	7	3°C	5°C
9	8	2°C	4°C
10	9	1°C	3°C
..	..		

Inference / Conclusion :

1. The water cools as the temperature in inner cylinder lowers due to freezing mixture and density of water increases. Therefore the temperature readings in both the thermometers are identical.

2. If temperature decreases below 4 °C water starts expanding and water becomes lighter due to decrease in density. It moves upward in the cylinder and the temperature of upper part of water column decreases to 0 °C.
3. The behaviour of water between the temperature 0 °C to 4 °C is known as anomalous.
4. What conclusion could you draw from the graph? How is anomalous behaviour of water proved? Initially the temperature T_1 decreases fast because of this mixture density increases and volume decreases. When the temperature reach 4 °C the volume of water increases and moves upwards and T_2 decreases.

Multiple Choice Questions

- What expands on reducing its temperature below 4 °C
 - 0
 - 4
 - 12
 - 5
- If two ice cubes are pressed against each other, they stick together firmly after some time because the melting point of ice reduce to below 0 °C due to the pressure applied.
 - Temperature of the ice cubes reduces.
 - the melting point of ice reduces to below 0 °C due to the pressure applied.
 - temperature around the ice cubes reduces.
 - the melting point of ice because 0 °C due to pressure.
- In above experiment the temperature of freezing mixture is 0 to 4 °C.
 - 0 to 4
 - 0 to -4
 - 10 to -20
 - 4 to 4
- In cold countries in winter, water carrying pipes sometimes break, because volume of water increases if the temperature is reduced below 0 °C.
 - volume of water reduces if the temperature is reduced below 0 °C.
 - air pressure increases.
 - the steel pipes become brittle due to reduced temperature below 0 °C.
 - volume of water increases if the temperature is reduced below 0 °C.
- Street ice cream vendors remove the water in their freezing mixture intermittently, as a result of which the temperature of the freezing mixture increases.
 - the temperature of the freezing mixture increases.
 - the temperature of the freezing mixture remains constant.
 - the temperature of the freezing mixture reduces further.
 - the weight of the freezing mixture reduces.



: Exercise :

- How to study the anomalous behaviour of water by using Hope's Apparatus?
During the study of anomalous behaviour of water using Hope's apparatus temperature T_1 and T_2 are recorded after every 30 seconds.
- Explain anomalous behaviour of water with the help of graph given on page 19.
The point of intersection at the two curves shows the temperature at which the density of water is maximum and volume is minimum. This temperature is 4 °C.
- What is the importance anomalous behaviour of water by using Hope's Apparatus?
The anomalous of water refers to the fact that water, unlike most substances expands when freeze the density of water increases as temperature decreases.

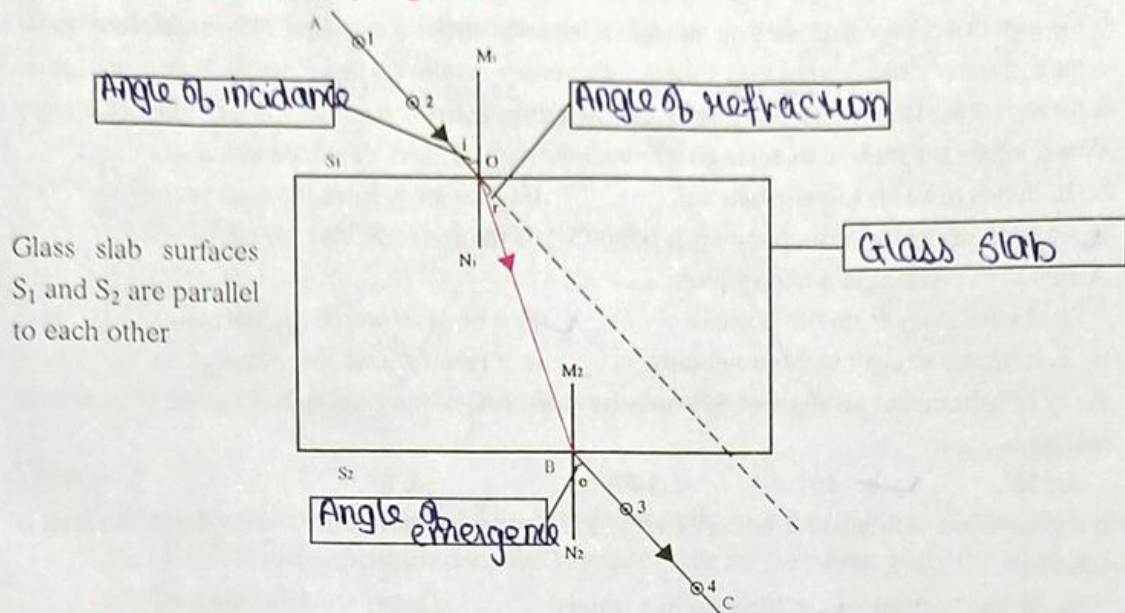
Remark and Signature

Practical No. 8

Aim : To verify the laws of refraction of light.

Apparatus : A glass slab, drawing board, drawing pins, paper pins, drawing sheet, etc.

Figure : (Label the following diagram.)



Procedure :

1. Take the drawing paper and fix it on the drawing board properly by using drawing pins.
2. Place the given glass slab at the center and draw its projections on the paper with a pencil.
3. Remove the slab and at point O, draw a normal M_1N_1 at the point O. Also, draw the ray AO at an angle of 30° with the normal.
4. Fix two paper pins 1 and 2 as the ray AO.
5. Now place the glass slab in its original place, look at the images of the paper pins 1 and 2 from the other side of the slab, and fix two more pins 3 and 4 in such a way that they are on the line joining the pins 1 and 2. The ray BC shows the emerging ray.
6. Remove the slab and join OB. Draw a normal M_2N_2 at the point B.
7. Measure the angle of incidence (i), the angle of refraction (r) and the angle of emergence (e).
8. Repeat the same procedure for 45° and 60° .

Observation :

Sr.No.	Angle of incidence (i)	Angle of refraction (r)	Angle of emergence (e)
1	30°	19° approx	30°
2	45°	26° approx	45°
3	60°	35° approx	60°

Inference / Conclusion :

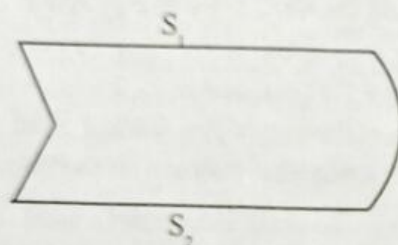
1. When light undergoes refraction through a glass slab, the ray and ray are parallel to each other.
2. The angle of incidence and the angle of emergence are of measures.

Multiple Choice Questions

- One student is doing experiment to draw the path of light rays passing through a glass slab. For this, he measured the angle of refraction and angle of emergence for every angle of incidence. In all the cases he found that.....
 - ☒ The angle of incidence is larger than the angle of refraction, but is almost equal to the angle of emergence.
 - ☐ the angle of incidence is smaller than the angle of refraction, but it is almost equal to the angle of emergence.
 - ☐ the angle of incidence is larger than the angle of refraction and also it is larger than the angle of emergence.
 - ☐ the angle of incidence is smaller than the angle of refraction, and also it is smaller than the angle of emergence.
- When a light ray makes an angle of 90° while entering a glass slab from air, it
 - ☐ bends towards the normal.
 - ☐ goes away from the normal.
 - ☒ goes straight without bending at point O.
 - ☐ returns back into the air.
- A light ray entering glass from water,
 - ☐ it goes away from the normal.
 - ☒ it bends towards the normal.
 - ☐ it travels straight without bending.
 - ☐ it returns back into water.
- A ray of light makes an angle of 50° with the surface S_1 of the glass slab. Its angle of incidence will be
 - ☐ 50°
 - ☒ 40°
 - ☐ 140°
 - ☐ 0°
- In a glass filled with water a coin placed at the bottom, if viewed in a skew angle the coin is found
 - ☐ at the bottom.
 - ☒ floating in a water.
 - ☐ bent.
 - ☐ broken.

: Exercise :

- If the glass slab is of the following shape, and the experiment is performed as above, will the effects be the same? Here, S_1 and S_2 are slab surfaces parallel to each other.



The incident ray bends towards the normal inside the glass slab and the refracted ray bends away from the normal. The extend at bending at light ray at the two opposite parallel and surface is equal.

- "During refraction of light through the glass slab, incident ray and emergent ray are parallel to each other." Explain.

The emergent ray is parallel to the incident ray but it is displaced slightly side ways.

- What would be the path of refracting ray and emergent ray, if two glass slab kept attached to each other.

If the two glass slabs are attached in parallel to each other and have same refractive index then the path difference will be as same as calculated for one slab. there will be no



Remark and Signature

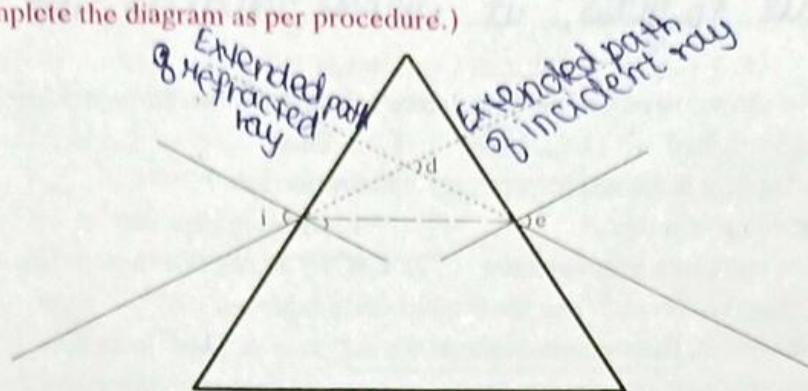
any deflection in 22 the path of light.

Practical No. 9

Aim : To trace the path of the rays of light through a glass prism and to find angle of deviation for different angles.

Apparatus : Prism, drawing board, a sheet of white paper, paper pins, drawing pins.

Figure : (Complete the diagram as per procedure.)



Procedure :

1. Take a sheet of white paper and fix it on the drawing board with the help of a drawing pins.
2. Place a glass prism along its triangular base at middle on the paper. Mark the boundary ABC of the glass prism.
3. Remove the prism and draw a normal LM through point Q on side AB.
4. Draw Ray PQ making an angle of 30° to the normal.
5. Fix two pins P_1 and P_2 on Ray PQ.
6. Place the glass prism back to its original position and look the images of pins P_1 and P_2 from the side AC.
7. Fix two more pins on P_3 and P_4 on this side such that pins P_3, P_4 and images of P_1, P_2 appear to be on a straight line.
8. Remove the prism and pins. Draw a line towards side AC through the points of pins P_3 and P_4 (S and R) which will intersect AC at point R.
9. Draw a normal on side AC through point R.
10. Extend Ray PQ and Ray SR on opposite side to intersect each other in point K.
11. Measure the angle of deviation d .
12. Repeat the procedure for angles of incidence 45° and 60° .

Observation

Observation No.	angle of incidence (i)	angle of refraction (r)	angle of deviation (d)
1	30°	19°	45°
2	45°	27°	35°
3	60°	60°	30°

Inference / Conclusion :

Initially, as the angle of incidence increases gradually, the angle of deviation decreases but after certain minimum value of incident angle, increases with the increase in the angle of incidence. The angle of incidence is greater than the angle of refraction. This means that a ray of light travels from air to glass, it bends towards the normal.

Multiple Choice Questions

1. Which colour shows maximum deviation when light is dispersed through prism?
☒ a. Violet b. Red c. Yellow d. Blue
2. How to convert light dispersed by prism again into white light?
a. By using white light rays. b. By using glass slab
c. By adding one more prism erected ☒ d. By adding one more prism inverted.
3. does not occur during the production of rainbow.
a. Refraction b. Dispersion c. Reflection ☒ d. Induction
4. We could observe path of light due to of light from tiny particles of solution of transparent medium.
☒ a. Scattering b. Dispersion c. Refraction d. Reflection
5. Dispersion of colour is minimum in visible range of light.
☒ a. Red b. Blue c. Green d. Violet

: Exercise :

1. Which colour deviate minimum during the dispersion of light through a prism?
wavelength is inversely proportional to the deviation in the path of the light. Red light suffers the least amount of deviation and violet the most.

2. If angle of incident increases angle of deviation decreases, but after a certain value of angle of incident, the angle of deviation increases. What is that specific value of angle of incident called? Does these angles same for light of all colour?

$$n_2 = \frac{\sin i}{\sin r}$$

for total internal reflection

i - critical angle

$$\therefore r = 90^\circ$$

Remark and Signature



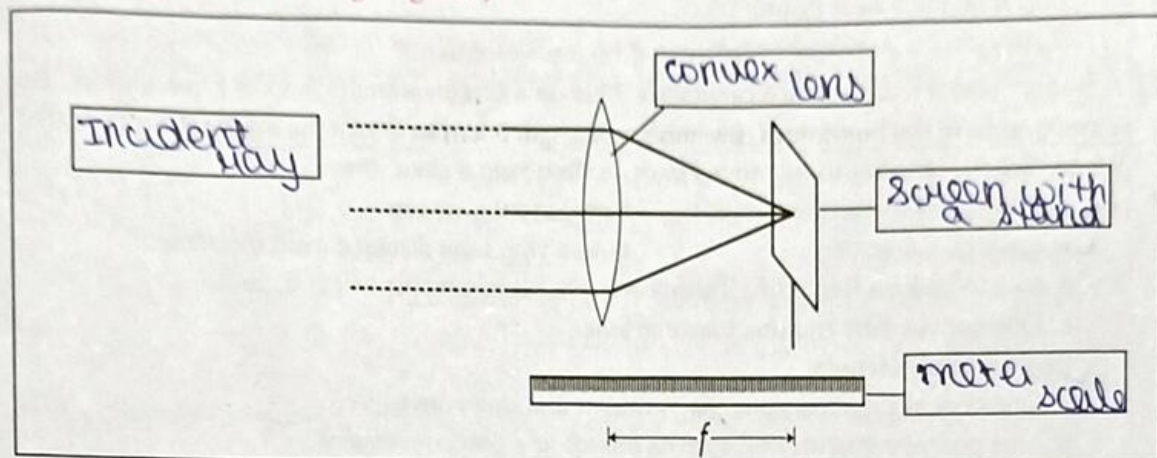
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Practical No. 10

Aim : To obtain the focal length of a convex lens.

Apparatus : Convex lens, lens holder, meter scale, screen with a stand, etc.

Figure : (Label the following diagram.)



Procedure :

1. Fit a convex lens in the lens holder.
2. Spot a distant object such as electric pole, a tree or a building.
3. Fit the screen on the stand and place it on the other side of the lens (opposite to object side.)
4. Move the screen back and forth to obtain a clear image of the chosen object on it.
5. Measure the distance between the lens and the screen.
6. Repeat the above steps two more times.
7. Now rotate the lens by 180° and repeat the above procedure.

Observation :

Least count of the meter scale = ... mm

Observation Table 1

Convex lens front surface facing the object

No.	Distant Object	Distance between lens center and screen
1	Tree	22.2
2	Tree	23.5
3	Tree	22.3
Average F1		$78.0 / 3 = 26.0$

Observation Table 2

Convex lens back surface facing the object

No.	Distant Object	Distance between lens center and screen
1	Tree	25.1
2	Tree	22.3
3	Tree	24.9
Average F2		$72.3 / 3 = 24.1$

Inference / Conclusion :

1. First focal length of the convex lens (F_1) ..25:0...cm
2. Second focal length of the convex lens (F_2).24:1....cm
3. From 1 and 2 above, is the lens used in this experiment a symmetric lens? (Yes / No)
If $F_1 = F_2$, then the lens is symmetric.

Multiple Choice Questions

1. A student obtain a clear image of window bars on a screen with the help of a convex lens. But in order to have the best image, his teacher suggested him to obtain the image of a distant tree. Where will the lens has to be moved in order to obtain a clear image?
a. away from the screen
b. behind the screen
☒ c. towards the screen
d. at a very long distance from the screen
2. If you want to make a Refracting Telescope, what components will you choose ?
a. One convex lens and one concave lens.
☒ b. Two convex lenses.
c. One concave mirror, one plane mirror and one convex lens.
d. One concave mirror, one convex mirror and one convex lens.
3. The image obtained while finding the focal length of convex lens is
a. a real and erect. b. virtual and erect. ☒ c. real and inverted. d. virtual and inverted .
4. For the same thickness of the lens, if the radius of curvature is increased, the focal length will
a. reduced. b. remain unchanged. ☒ c. increased. d. be imposible to estimate.
5. At what distance a watch maker must hold his lens from the watch ?
☒ a. at the focal length b. at less than focal length
c. at more than the focal length d. at zero distance

: Exercise :

1. Explain the working of compound microscope by focal length of convex lens.

The simplest compound microscope is constructed from two convex lenses the objective lens of short focal length and eye piece has a bigger focal length. Higher magnification can be obtained by combined effect of lenses.

2. When would virtual image produced by convex lens? Why doesn't virtual image drawn on a screen?

Virtual image can not be drawn on an screen because of the technology that people have to put the material in the contact lenses.

3. Give your opinion about radius of curvature of a convex lens having different F_1 and F_2 ? What would the lens called having equal F_1 and F_2 ?

The distance between optical centre and centre of curvature is called radius of curvature.



Remark and Signature

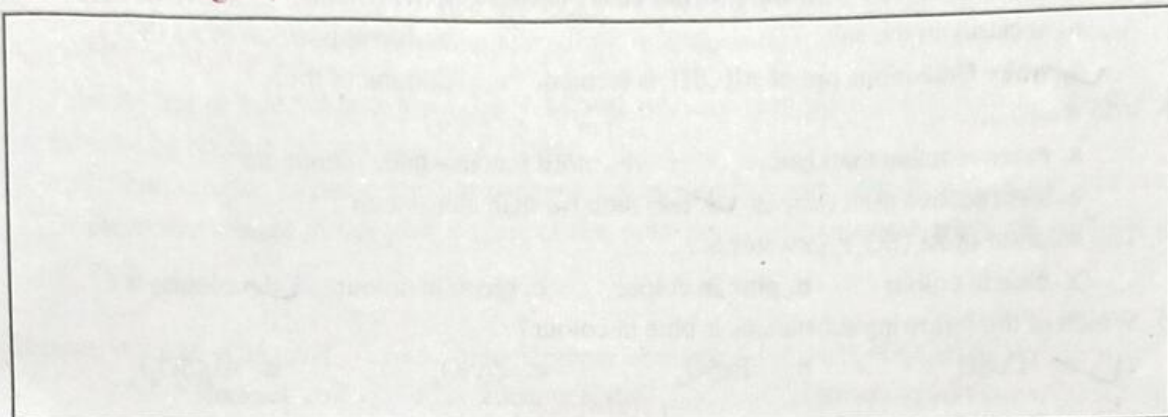
Practical No. 11

Aim : To study the reactions of the metals Zn, Fe, Cu and Al with the solutions of salts ZnSO_4 , FeSO_4 , CuSO_4 , $\text{Al}_2(\text{SO}_4)_3$ and to arrange these metals in the decreasing order of their reactivity.

Apparatus : Test tubes, test tube stand, distilled water, sand paper, etc.

Chemicals : The metals Zn, Fe, Cu and Al; solutions of the salts ZnSO_4 , FeSO_4 , CuSO_4 , $\text{Al}_2(\text{SO}_4)_3$

Figure : (Draw figure)



Procedure :

1. Clean all the metals with sand paper and cut them into small pieces.
2. Take 10ml of samples of each of the solutions of ZnSO_4 , FeSO_4 , CuSO_4 , $\text{Al}_2(\text{SO}_4)_3$ in separate test tubes.
3. Put one or two pieces of aluminium metal in each test tube. See what happens and note the observations.
4. Repeat the same procedure for the remaining three metals.

Observations :

Metal	Solution			
	$\text{Al}_2(\text{SO}_4)_3$	ZnSO_4	FeSO_4	CuSO_4
Aluminium	--	Zinc is displaced	Iron is displaced	Copper is displaced
Zinc	No Reaction	--	Iron is displaced	Copper is displaced
Iron	No Reaction	No Reaction	--	Copper is displaced
Copper	No Reaction	No Reaction	No Reaction	--

Inference / Conclusion :

The decreasing order of reactivity of the metal is $\text{Al} > \text{Zn} > \text{Fe} > \text{Cu}$

Multiple Choice Questions

- The proper procedure of the experiment to show that zinc is more reactive than copper is
 - ☒ Prepare a copper sulphate solution and immerse a copper plate in it.
 - Prepare zinc sulphate solution and immerse a copper plate in it.
 - Heat zinc and copper plate.
 - Add dilute nitric acid to both the plates.
- The solution of $\text{Al}_2(\text{SO}_4)_3$ in water is not clear, because of
 - impurity in the salt.
 - decomposition of $\text{Al}_2(\text{SO}_4)_3$.
 - ☒ white Gelatinous ppt of $\text{Al}(\text{OH})_3$ is formed.
 - none of these.
- Iron is
 - more reactive than zinc.
 - more reactive than aluminium
 - less reactive than copper
 - ☒ less reactive than aluminium
- The solution of $\text{Al}_2(\text{SO}_4)_3$ in water is
 - blue in colour
 - pink in colour
 - green in colour
 - ☒ colourless
- Which of the following substances is blue in colour?
 - ☒ CuSO_4
 - FeSO_4
 - ZnSO_4
 - $\text{Al}_2(\text{SO}_4)_3$

: Exercise :

- Why do all the metal pieces get polished before the experiment?
 ① To remove the layer of metal oxide and to prevent rusting of iron metal pieces get polished before the experiment.
- Why can not each metal react to its own salt (Eg. Cu and CuSO_4)?
 Each metal does not react to the own salt because then there will be no displacement. As it is known that accordingly to reactivity more reactive element displaces the less reactive element.
- Arrange the non metals iodine, fluorine, bromine, chlorine in decreasing order of their reactivity?
 Fluorine
 Chlorine
 Bromine
 Iodine



Remark and Signature

Practical No. 12

Aim : To study oxidation and addition reactions of carbon compounds.

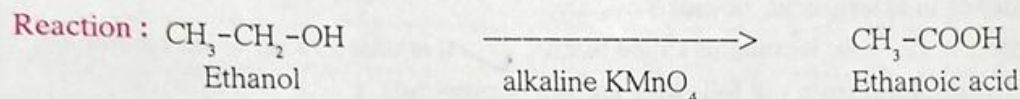
Apparatus : Test tubes, dropper, burner, etc.

Chemical substances : Ethanol, dilute solution of sodium carbonate, dilute solution of potassium permanganate, alcoholic solution of iodine (tincture iodine), liquified vanaspati ghee, various vegetable oils (groundnut oil, safflower oil, sunflower oil, etc.)

Procedure :

A. Oxidation of ethanol

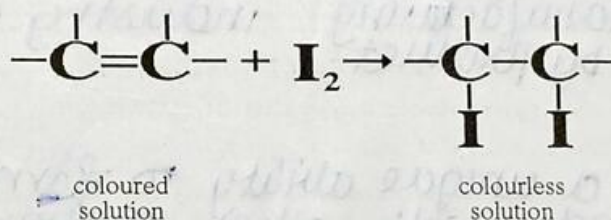
1. Take 2-3ml of ethanol in a test tube. Add 5ml sodium carbonate solution to it. Warm the mixture by holding the test tube on a burner.
2. Add dilute solution of potassium permanganate dropwise in it and keep on stirring the mixture.
3. Observe the change in the pink colour of the potassium permanganate when its addition is started.



B. Addition reaction of fatty acids

1. Take 2ml oil in a test tube. Add 4 drops of tincture iodine in it and stir.
2. Observe whether original colour of iodine disappears or not.
3. Repeat the same procedure using the other oils and vanaspati ghee and note in the observation table.

Reaction :



Observations :

- A. i. When a dilute solution of potassium permanganate is added to ethanol, initially the pink colour disappears.
- ii. When the addition is continued further, the pink colour of potassium permanganate does not vanish and stays there.

B.

Oil sample	Colour change observed in the solution
groundnut oil	Pink colour vanishes
safflower oil	pink colour vanishes
sunflower oil	pink colour vanishes
vanaspati ghee	Pink colour does not vanishes.

Conclusion / Inference :

- a. Potassium permanganate oxidizes ethanol to ~~ethanoic~~ acetic. As potassium permanganate is consumed in this reaction its pink colour vanishes.
- b. Iodine is consumed due to its addition to fatty acid. Therefore the coloured solution becomes colourless. But when the same procedure is followed for vanaspathi ghee, a similar colour change is not observed. As vanaspathi ghee is saturated hydrocarbon the addition reaction does not occur there.

Multiple Choice Questions

- What type of reaction is the transformation of ethanol into ethanoic acid?
a. Addition reaction ☐ b. Substitution reaction ☒
c. Oxidation reaction ☒ d. Dehydration reaction ☐
- In which of the following substances, is ethanol not used?
a. tonics b. cough mixture ☒ c. candle d. alcohol
- What type of carbon-carbon bonds are present in vanaspathi ghee?
a. single bonds ☒ b. double bonds c. triple bonds d. none of these
- Iodine decolourises in stearic acid, because
a. it is saturated. b. it contains single bonds. ☒ c. it is unsaturated. d. cannot tell.
- The saturated hydrocarbon from the following carbon compounds is
a. Ethene b. Ethyne ☒ c. Ethane d. Benzene

: Exercise :

- In which industry ethanol is obtained as a byproduct?
Ethanol being good solvent it is used in medicines such as cough mixture, cough mixture etc. In sugar manufacturing industry ethanol is obtained as by product.
- What is Catenation?
Carbon has a unique ability to form strong covalent bonds with other carbon atoms. This results in formation of big molecules. This property of carbon is called catenation.
- Why is it made compulsory to mix ethanol in a fuel?
Ethanol is a renewable solvent and is the by product of sugar industry. The reason ethanol is mixed with fuel is because it burns itself clean resulting in a decrease in the carbon monoxide, CO_2 and hydrocarbons which

Remark and Signature

Reduces air pollution



Practical No. 1

Aim : To study mitosis and meiosis.

Apparatus : Permanent slides of mitosis and meiosis, compound microscope.

Procedure :

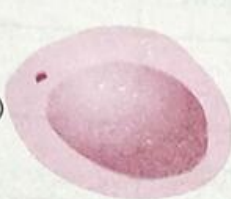
1. Observe the permanent slides showing different stages of mitosis, under compound microscope.
2. Repeat the same procedure for meiosis also.
3. Label the following diagrams:

Diagrams: (Label the following diagram.)

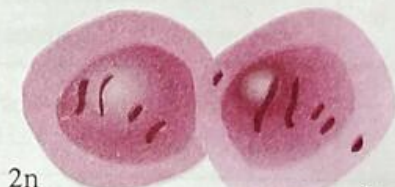
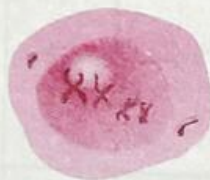
Stages of mitosis :

one Diploid cell

46 Chromosomes ($2n$)



DNA Model formation $4n$



Prophase

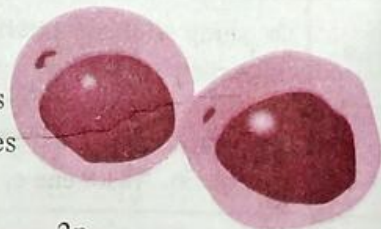
Metaphase

Anaphase

Telophase

Two Diploid cells

46 Chromosomes



$2n$

$2n$

Stages of meiosis :



1

2

3



4

5

Metaphase-I

Anaphase-I

Telophase-I

Two singular Cells

Observations :

Main differences between mitosis and meiosis:

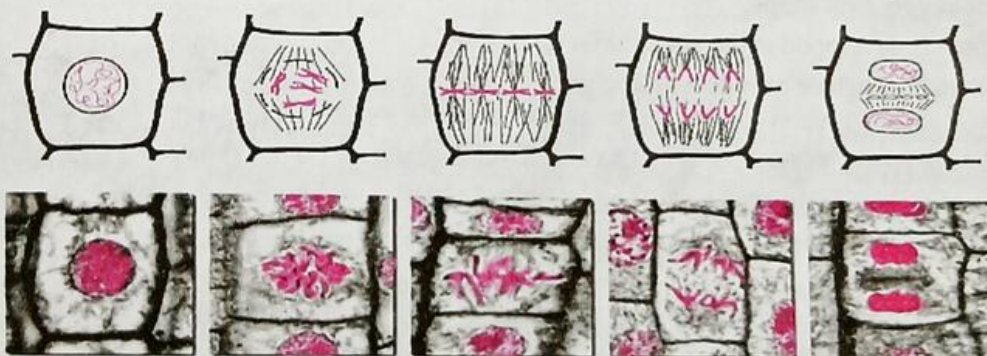
- | | |
|--|------------------------------------|
| 1. Mitosis occurs in somatic and ^{somatic cell} | Meiosis occurs in germ cells |
| 2. One cell gives two daughter cells | One cell gives four daughter cells |
| 3. Diploid cells are formed | Haploid cells are formed |
| 4. Prophase is not lengthy | Prophase is very lengthy |

Inference :

- 1) Due to mitosis the number of chromosomes in the cell ~~remains same~~ ^{remains same} therefore, this division occurs in ~~somatic~~ ^{somatic} cells.
- 2) Due to meiosis the number of chromosomes in the cell ~~reduce by half~~ ^{reduce by half}, so this division occurs in ~~diploid~~ ^{diploid} cells.

Multiple Choice Questions

1. A diploid cell means
a. n ☒ b. $2n$ c. $3n$ d. $4n$
2. In this stage of mitosis, the daughter chromosomes appear like bunch of bananas.
a. Prophase b. Metaphase ☒ c. Anaphase d. Telophase
3. During metaphase this change occurs in the chromosomes
a. get coil shape. ☒ b. arranged parallel to equatorial plane.
c. get destroyed due to breaking d. get folded.
4. How many times a cell will have to divide mitotically, to form 128 cells?
☒ a. 7 b. 14 c. 28 d. 32
5. Suppose, you have supplied with onion root to observe steps in cell division. If you want to count the number of chromosomes, which step will you observe?
a. Prophase ☒ b. Metaphase c. Anaphase d. Telophase



: Exercise :

1. What would have happened if meiosis not occur during the production of human sperm cell and egg cell ?

If meiosis does not take place in the gametes then there will be $46 + 46 = 92$ chromosomes in zygote. Such chromosomes number will produce large scale abnormalities.

Remark and Signature

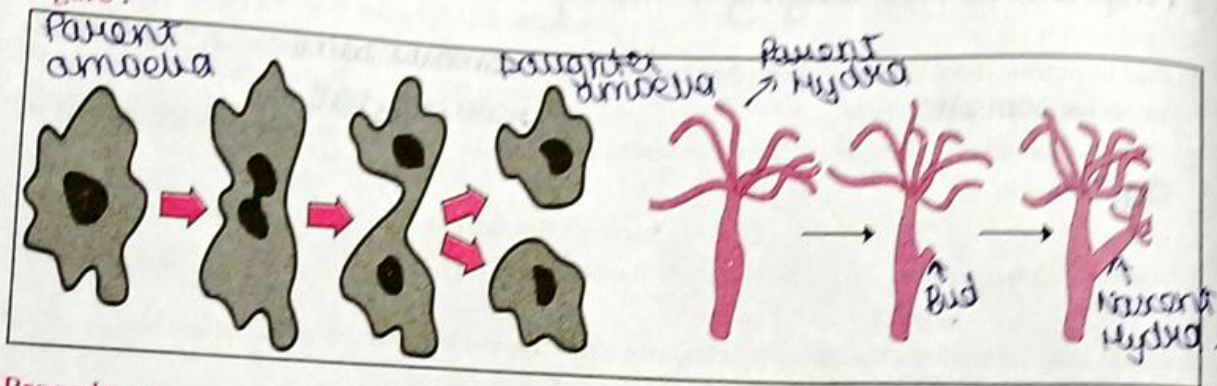


Practical No-2

Aim : To study binary fission in Amoeba and budding in Hydra with the help of permanent slides.

Apparatus : Compound microscopes, permanent slides showing binary fission in Amoeba and budding in Hydra.

Figure :



Procedure :

1. Take the permanent slides showing binary fission in Amoeba and budding in Hydra.
2. Adjust eyepiece and mirror of compound microscope. Observe each slide first under low power and then under high power objective lens.
3. Note down your observations.

Observation :

A. Binary fission in Amoeba :

1. At the beginning, parent cell of Amoeba gets elongated.
2. Nucleus gets oval shape.
3. A notch is developed at the site of division.
4. Two small daughter cells of Amoeba are formed.

B. Budding in Hydra :

1. A small outgrowth is seen on the body of Hydra.
2. It becomes multicellular due to enough growth.
3. This outgrowth detaches in the form of bud having mouth and tentacles.

Inference / Conclusion :

1. Amoeba and Hydra reproduce by asexual method.

Multiple Choice Questions

1. Previously Raju observe Hydra bud under low power objective lens. But afterwards he used high power to observe
a. micro parts in small space. b. maximum space around.
c. maximum of bud. ☒ d. All of the above.

2. Which of the following is asexual method of reproduction?
 a. Division b. Budding c. Spore formation ☒ d. All of these.
3. To observe Amoeba and Paramecium in a water drop, the drop was taken on a glass slide. If both the organisms were clearly seen, then the stain used must be
☒ a. Methylene blue. b. Iodine. c. Safranin. d. Eosin.
4. is the type of reproduction in Hydra.
 a. Binary fission ☒ b. Budding c. Multiple division d. None of these.
5. While observing permanent slide of Amoeba, Ramesh saw oval shaped nucleus. This nucleus indicates
☒ a. binary fission b. budding c. multiple division d. none of these.

: Exercise :

1. Explain with examples - different methods of asexual reproduction in plants.

1] Budding - Hydra

2] Vegetative propagation - potatoes, radish.

2. Name the process where cytoplasmic material is divided into daughter cells.

Cytokinesis is the process where cytoplasmic material is divided into daughter cells.

3. What changes occurs in Amoeba to survive in adverse conditions?

Whenever there is lack of food or any other adverse condition many amoebas survive by encystment. The amoeba forms protective covering around their body called cyst and remain encysted. Inside the cyst by repeated nuclear and cytoplasmic division many amoebulae are formed. After divide of favourable condition, cyst breaks and amoebulae released.

4. How does nutrition takes place in bud of Hydra?

In the case of hydra, under favourable condition an outgrowth is formed called bud. Bermal layer and digestive cavity of budding hydra are continuity with the parent hydra. Parent hydra supplies nutrition to the budding hydra. Budding hydra separates from parent hydra after proper growth.

Remark and Signature

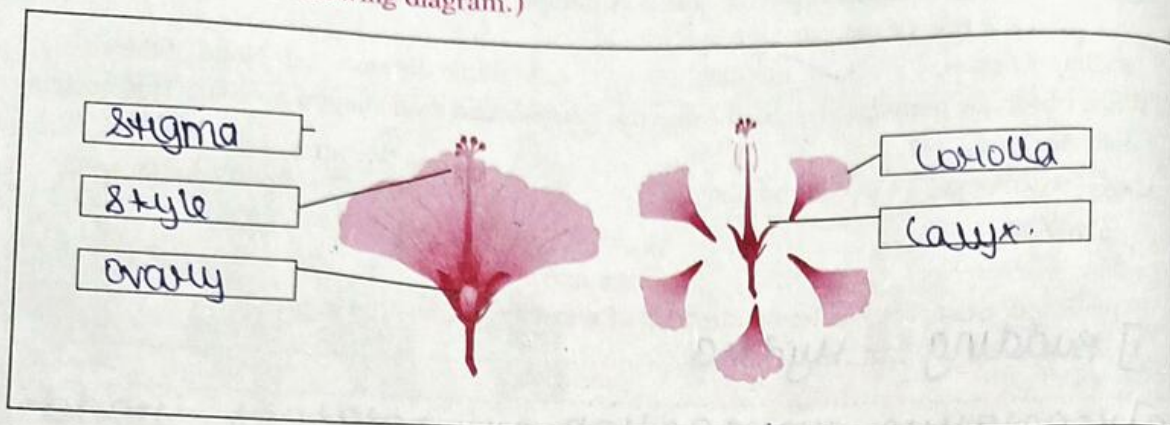


Practical No. 3

Aim : To study the reproductive system in Hibiscus flower.

Apparatus : Hibiscus flower, forcep, blade, simple microscope, etc.

Figure : (Label the following diagram.)



Procedure :

1. Observe the Hibiscus flower and note down its whorls.
2. Separate androecium and gynoecium by removing sepals and petals.
3. Take a longitudinal section from stigma to ovary with the blade.
4. Observe anthers in the same way.

Observation :

Whorl	Number of components	Structure	Function
Calyx	Five	Green sepals	To protect inner whorls
Corolla	Five	colourful petals	To attract insect for pollination
Androecium	Innumerable stamens	Anthers	To produce pollen grains
Gynoecium	Five carpel	encloses the ovules	To produce female gametes of flower.

Inference / Conclusion :

1. ~~Androecium~~ from flower is male whorl, while ~~Gynoecium~~ is female whorl.
2. Male gamete from pollen grain and egg cell from ovule, these two haploid cells unite to form a diploid zygote. Thus the process of seed and fruit formation begins.
3. Flower is the reproductive organ of plants.

Multiple Choice Questions

1. and are the essential whorl.
a. Calyx b. Corolla ✓ c. Androecium ✓ d. Gynoecium
2. is not unisexual flower.
a. Coconut b. Papaya ✓ c. Gulmohar d. Water-melon
3. The process of union of male gamete with egg cell is called
✓ a. fertilization. b. pollution. c. double fertilization. d. germination.
4. Second male gamete unites with two polar nuclei to form
a. zygote. b. embryo. c. embryo sac. ✓ d. endosperm.
5. After fertilization develops into seed and into fruit.
a. endosperm ✓ b. ovary c. embryo sac ✓ d. egg cell

: Exercise :

1. Seeds of some plants don't germinate. How are their next generation formed?

Seed germination is not only process of reproduction in plants. Plants can also reproduce by other method like vegetative propagation, budding etc. Hence, seeds of some plants don't germinate but then also their next generation is formed.

2. What are the artificial methods of vegetative reproduction in plants? Give examples.

There are various artificial method of vegetative reproduction in plants.

- 1] Cutting - In this a portion of a plant is cut and planted in the soil, usually a leaf or a stem.
- 2] Grafting - In this, the cutting from some other plant is added to the plant stem embedded in soil.

3. Why does hibiscus not bears fruits even though androecium and gynoecium present in it?

Hibiscus is a plant that does not produce any fruit or we can say that the fruit is hidden. The fruit of hibiscus flower is the five lobed of pods contain seeds like any other fruit. So the hibiscus plant doesn't have any distinct fruit even though androecium and gynoecium are present in it.

4. Why do androecium and gynoecium of same flower matures at different time?

Androecium and gynoecium of same flower matures at different time to avoid self pollination.

Remark and Signature



Practical No. 4

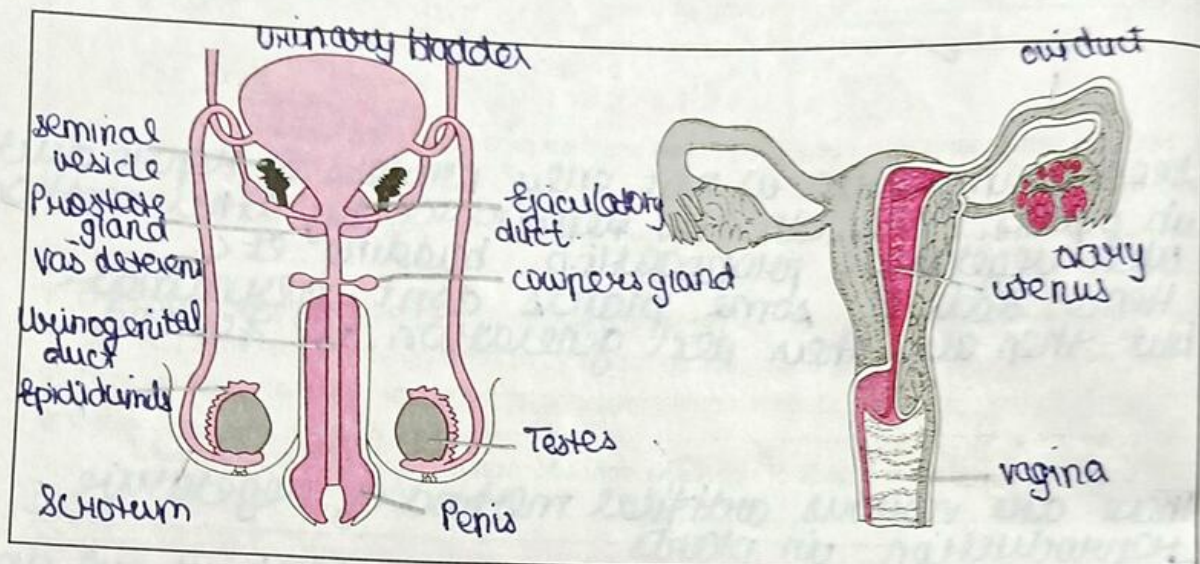
Aim : To study human reproductive systems.

Apparatus : Models of male reproductive system and female reproductive systems.

Procedure :

1. Observe the models of male reproductive system and female reproductive systems.
2. Note down different organs found in them.
3. Find out their functions and note down.
4. Label the diagrams given below.

Figure :



Observations :

Sr.	Name of organ in Male Reproductive System	Function
1	Testis	Production of sperms and testosterone
2	Scrotum	Protection and temperature control of testis
3	Seminal Vesicles	Secretion of seminal fluid
4	Penis	Transferring sperms to vagina and urination.

Sr.	Name of organ in Female Reproductive System	Function
1	Ovary	Production of sperms and estrogen and progesterone hormones like
2	uterus	Growth and development of foetus
3	Vagina	Birth canal passage for menstruation
4	Oviduct	Passage to transport ovum to uterus.

Inference / Conclusion :

1. Male reproductive system produces haploid gametes called sperm.
2. female reproductive system produces haploid (n) gametes called ovum.
3. Sperm and ovum unite to form a diploid (2n) zygote and develops a human embryo.

Multiple Choice Questions

1. produces sperm cells.
a. Seminal vesicle
b. Urinary bladder ✓
c. Testis ✓
d. Ampulla of duct
2. The length of a sperm is micrometer.
a. 400
b. 5
c. 60 ✓
d. 600
3. At the time of birth, a female has ova in her ovary.
a. 2 to 3 lakh
b. 10 to 20 lakh
c. 2 to 4 lakh ✓
d. 20 to 40 lakh
4. In context to menstrual cycle, which is the wrong statement?
✓ a. Around 40ml of blood is discarded in a menstrual cycle.
b. The menstrual fluid including blood, easily coagulates.
c. Menstrual cycle generally starts from 12th year of age.
d. Disturbing menstrual cycle with hormonal tablets, causes side effects.
5. are the ways of family planning.
a. To take contraceptive tablets.
b. To use Nirodh (Condoms).
c. To install copper -T.
✓ d. All of these.

Exercise :

1. How do the characteristics of mother and father get transmitted in a human embryo?
In humans, haploid male and female gametes are united to form diploid zygote. Zygote carries characteristics of both parents which are transmitted via gametes. In gametes, due to meiosis process, recombination occurs. This is formed of 23 chromosomes transmitted in human embryo.
2. Give the names of bisexual animals.

Bisexual animals - Starfish, earthworm, deep sea squid etc.

3. Why male reproductive system is called as urinogenital system?

The word urinogenital is composed of two words urine and genital. Hence, urinogenital system refers to the parts of both urinary system as well as reproductive system. In male reproductive system there is only one opening for both urine and semen. Therefore male reproductive system is called urinogenital system.

Remark and Signature



Practical No. 5

Aim : Comparative study of ecosystems in the surroundings.

Apparatus : Note book, pen, pencil, old newspapers etc.

Procedure :

1. Select any two different ecosystems in 1 SqKm area around your home or school. Eg. pond, meadow, garden, large trees like banyan, peepal, tamarind, mango, agricultural farm etc.
2. Note down abiotic factors found in them.
3. Collect the samples of plants with different sizes and types. Keep seedlings or branches of plants in folds of old newspapers and dry them properly. Make a scrap book from these herbarium sheets.
4. Note down the names of animals found in both the ecosystems. Collect their diagrams/photos.
5. Find out the differences between the biotic factors in both the ecosystems. Find out the reasons.

Observations :

Sr.	Name of the ecosystem	List of biotic factors	List of abiotic factors
1	Aquatic ecosystem	aquatic animal, fishes	water, sunlight
2	Grassland ecosystem	organic matter	climate soil
3	Tundra ecosystem	various animal and plant	snow, climate
4	Desert ecosystem	cactus, camel etc.	sunlight, water shortage

Other important notes :

1. Ecosystem varies from small pond to the huge earth.
2. Ecosystem is formed by interaction between various biotic and abiotic factors present in environment.
3. There is ecosystem diversity on the earth.

Inference / Conclusion

- 1] An ecosystem is formed due to interaction between biotic / abiotic factors.
- 2] Ecosystem can be natural or artificial.
- 3] Earth is the largest ecosystem.
- 4] Each region contain different types of ecosystem.

Multiple Choice Questions

1. Butterfly is a consumer.
☒ a. primary b. secondary c. tertiary d. none of these
2. of the following is omnivorous.
 a. Cow b. Grasshopper c. Worm ☒ d. Sparrow
3. Food web contains many
 a. primary consumers. b. secondary consumers. c. tertiary consumers. ☒ d. food chains.
4. Autotrophic plants are at the level of food chain.
 a. tertiary nutrition b. secondary nutrition ☒ c. producers d. highest
5. Man is animals.
 a. herbivorous b. carnivorous c. autotrophic ☒ d. omnivorous

: Exercise :

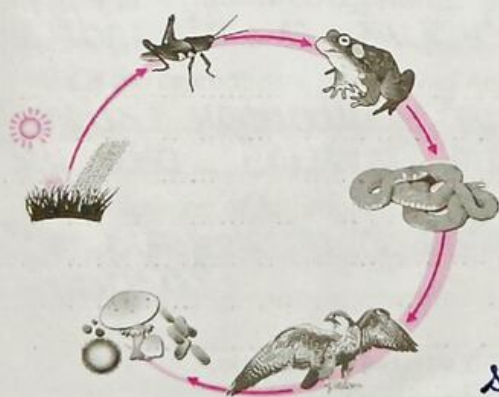
1. Find out in the past if there were some biotic and abiotic factors in the ecosystems you have selected. But now they are not found. If this is the situation, find out the reasons for their diminution.

Yes there were some biotic and abiotic factors in the ecosystem selected. Biotic factors such as some plants and animals extinct because they didn't adapt with their surrounding. Abiotic factors includes non-living matter such as soil, water, nutrients etc. Increasing population and pollution are the reasons behind why they diminished.

2. Did you find any examples of human interference in these ecosystems?

Yes due to various human activities cycle of environment is disturbing such as pollution, population explosion, industrialization causing great impact on ecosystem.

3. What is the relation with ecosystem of figure shown below?



The animals shown in the figure are depend on each other for survival. Producers are very important in any type of ecosystem. To maintain environmental balance each animal is important. This figure shows the food chain.

Remark and Signature



Practical No. 6

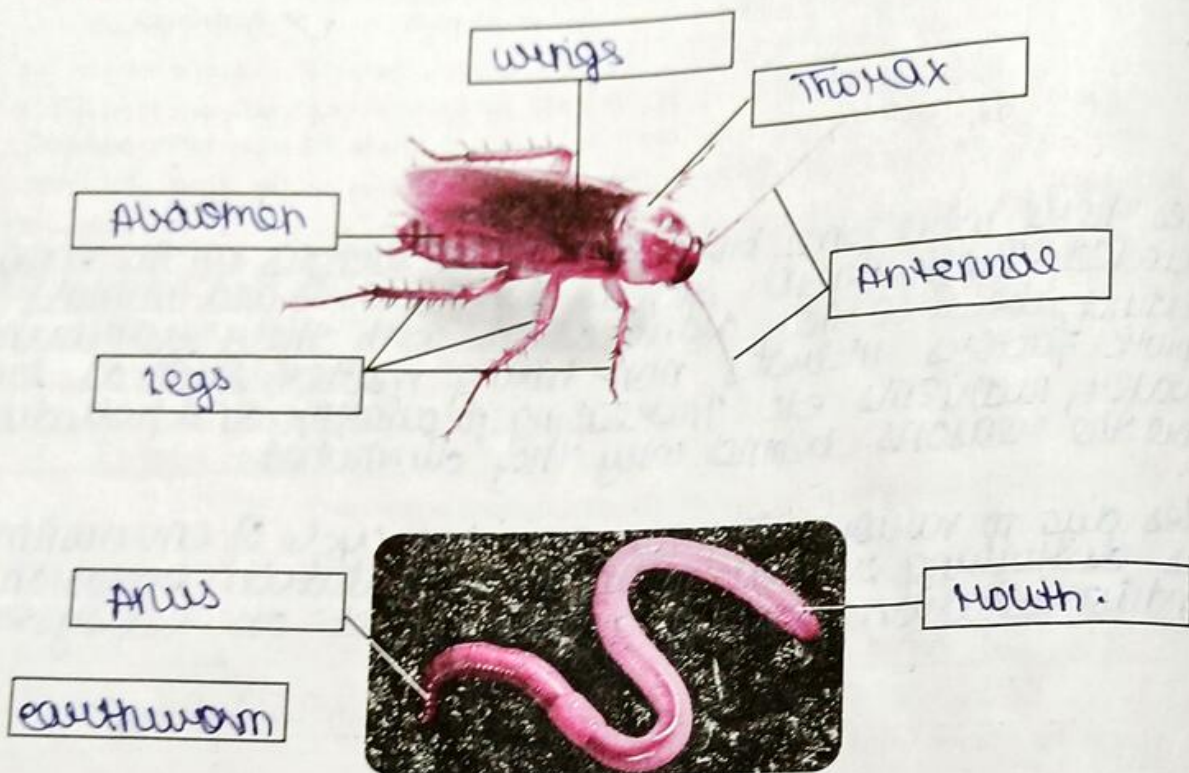
Aim : To study the characteristics of non-chordate animals a. Earthworm b. Cockroach.

Apparatus : Live or preserved specimens of earthworm and cockroach.

Procedure :

1. Observe the given specimens according to the points given below and study their characteristics :
 - a. A prominent character of its phylum.
 - b. Label the following diagram.

Figure :



Observation :

a. Earthworm

Kingdom	- Animalia
Sub-kingdom	- Non-chordata
Phylum	- Annelida
Example	- Earthworm

Characteristics :

1. Long cylindrical and metamerically segmented body covered with cuticle.
2. Its called 'Friend of farmer' as it dwells the land and makes it loose.

b. Cockroach

Kingdom	- Animalia
Sub-kingdom	- Non-chordata
Phylum	- Arthropoda
Class	- Insecta
Example	- Cockroach

Characteristics :

1. Cockroach is a terrestrial and nocturnal insects.
2. Head, thorax and abdomen are the three main body parts. Cockroach has three pairs of jointed legs and two pairs of wings.

Multiple Choice Questions

1. If non labelled diagram of earthworm is given, then which special character must be shown to include it in a specific phylum?
a. eye ☒ b. metamerical segmentation c. clitellum d. anus
2. Due to which similar characteristic honey bee and cockroach are included in the same phylum?
a. wings b. three pairs of legs ☒ c. jointed appendages d. antenna
3. Which of the following characters are not found in Cockroach?
a. compound eyes b. two pairs of wings
☒ c. no jointed appendages d. three pairs of legs with jointed appendages
4. Heart of cockroach has compartments.
a. 2 b. 3 c. 4 ☒ d. 13
5. Earthworm has around segments on its body.
a. 60 to 100 ☒ b. 100 to 120 c. 120 to 140 d. more than 140

: Exercise :

1. Make a list of another animals in the phylum Annelida.

Animals in phylum Annelida - leech, Nereis

2. What are the organs of locomotion in Annelids?

Organs of locomotion in Annelids are setae, parapodia or suckers.

3. What is the reproductive speciality of Annelids?

Annelis reproduce by sexually reproduction method. They are unisexual or hermaphrodites.

4. The exoskeleton of Arthropods is made up from which material?

The exoskeleton of Arthropods is made up of chitinous material.

Remark and Signature



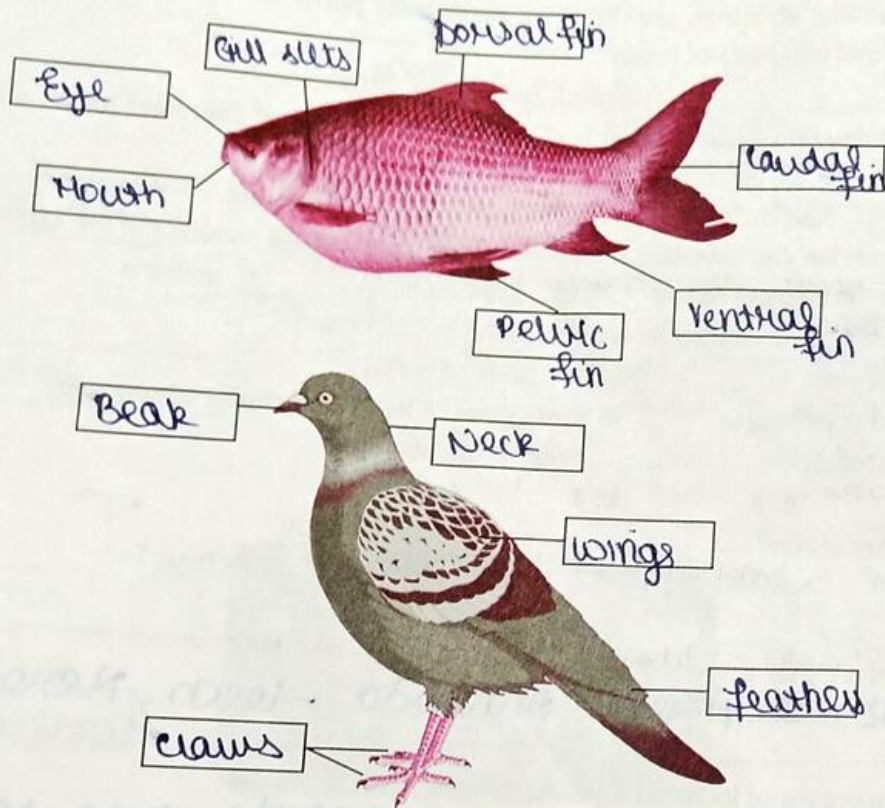
Practical No. 7

Aim : To study the characteristics of chordate animals a. Fish b. Pigeon

Apparatus : Live or preserved specimens of fish and pigeon.

Procedure :

1. Observe the given specimens according to the points given below
 - a. A prominent character of its phylum.
 - b. Changes with reference to the adaptation.
2. Label the diagrams shown below.



Observation :

a. Fish :

Kingdom	- Animalia
Phylum	- Chordata
Sub-phylum	- Vertebrata
Class	- Pisces
Example	- Rohu fish

Characteristics :

1. Fish is a cold blooded animal with a spindle shaped body to minimize water-resistance.
2. Fish breaths with gills.

b. Pigeon :

Kingdom	- Animalia
Phylum	- Chordata
Sub-phylum	- Vertebrata
Class	- Aves
Examples	- Pigeons

Characteristics :

1. Pigeons body is covered with feathers. Forelimbs are modified into wings. Heart has four compartments.
2. Pigeon has hollow bones and air bags inside the body. This is aerial adaption.

Multiple Choice Questions

1. Fish is blooded animal.
a. mix ☒ b. cold c. warm d. None of above
2. Pigeon's heart has compartments.
a. 1 b. 2 c. 3 ☒ d. 4
3. Due to which similar character, birds and fishes are in the same phylum?
a. tapering head b. heavy chest
c. caudal fin ☒ d. Vertebral column
4. The caudal fin of fish and pigeon is useful to
☒ a. increase the velocity. b. decrease the velocity.
c. change direction. d. maintain balance.
5. Bird's are modified into wings.
☒ a. forelimbs b. hindlimbs c. thorax parts d. non of these

: Exercise :

1. Explain the adaptation in fish body which makes floating on water as well as going in deep water easy.

Fishes have gills to breathe the dissolved oxygen in deep water. They have scaly body and fins to float on the water.

2. Which factor decides whether the animal is 'cold blooded' or 'warm blooded'?

The internal body temperature decides whether the animal is cold blooded or warm blooded.

3. What are the changes in birds bodies that adapt them to aerial habitat?

The body of birds are covered with feathers which help them to fly. Their forelimbs are modified into wings which help them to maintain their speed. These changes in the birds bodies adapt them to aerial.

Remark and Signature



Practical No. 8

Aim : To study the physical and chemical properties of acetic (ethanoic) acid.

Apparatus : Test tubes, test tube stands etc.

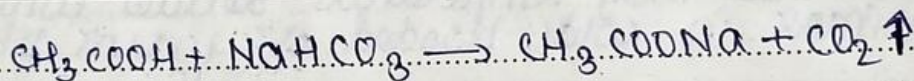
Chemicals : Acetic acid, red and blue litmus solutions, sodium bicarbonate.

Procedure :

1. Take 2-3ml of acetic acid in a clean, dry test tube. Note down its colour and odour.
2. Add 4-5ml water to the acetic acid in the test tube and mix well.
3. Take 2ml of acetic acid in a test tube and add blue litmus solution to it. Do the same procedure for red litmus solution. Observe both the test tubes.
4. Take 2 ml of acetic acid in a test tube and add a pinch of sodium bicarbonate. Observe.

Observation and Inference / Conclusion :

Sr.No	Test	Observations	Inference
1	Odour	Odour like Vinegar	Acetic acid has pungent like odour.
2	Solubility in water	Acetic acid ...soluble... in water.	Acetic acid is soluble in water.
3	Effect on litmus		
a.	Blue litmus	Turns red	Acetic acid is acidic in nature.
b.	Red litmus	Does not change colour	
4	a. Reaction with sodium bicarbonate	a. Colourless and odourless gas bubbles are evolved	CO ₂ gas is evolved when acetic acid reacts with sodium bicarbonate.
	b. Pass this gas through lime water.	b. At first lime water turns milky, but if the gas is allowed to pass for some more time, the lime water becomes colourless again.	



Multiple Choice Questions

1. Vinegar used as preservative for pickles, is % aqueous solution of acetic acid.
a. 1 to 2 ☒ b. 5 to 8 c. 25 to 30 d. 100
2. Acetic acid looks like ice due to freezing at °C.
a. 0 b. -10 ☒ c. 17 d. 4
3. Acetic acid is
a. yellowish. ☒ b. colourless. c. white. d. pink.
4. The functional group in ethanoic acid is
a. alcohol. b. aldehyde. c. ketone. ☒ d. carboxylic.
5. 2ml ethanoic acid was taken in test tubes - A, B and C. 2ml, 4ml and 8ml of water was added to the test tubes A, B and C respectively. Which test tube will show clear solution?
a. only A b. only B c. only A and B ☒ d. all

: Exercise :

1. Which microbe is used to extract ethanol from fruit juices and sugar molasses?

Saccharomyces cerevisiae is the microbe used to extract ethanol from fruit juice and sugar molasses.

2. Which main product is obtained by microbial fermentation as ethanol?

A microbial fermentation process in which sugar is converted into ethanol is called fermentation. In this, sugars like glucose, fructose, sucrose are converted into ethanol and CO₂ organic substances.

3. Give examples of beverages and chemicals obtained by microbial fermentation of different organic substance.

Example of beverages and chemicals obtained by microbial fermentation of different organic substance are as follows: apple cider, wine, cellulose, ethanol, acetic acid etc.

4. Why the Study/ Checking chemical and physical properties of daily used substances/ material is essential?

The knowledge of physical and chemical properties of daily use substance is necessary to optimally take its advantage for its storage as some substances lose their shelf life at high temperature in order to avoid further complication like allergies etc.

Remark and Signature



Practical No. 9

Aim : To study the characteristics of microbes that are industrially useful.

Apparatus : Compound microscope and permanent slides of any of the following microbes - streptococcus thermophilus, lactobacillus acidophilus, saccharomyces cerevisiae, acetobacter, aspergillus oryzae.

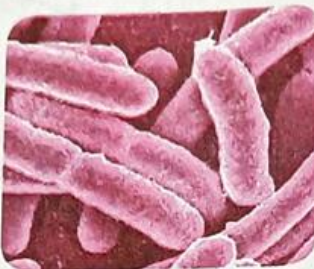
Procedure :

1. Select any four permanent slides among the above available in your laboratory. Observe them under compound microscope and note their characteristics.
2. Find out industrial application / use of each.

Figure :



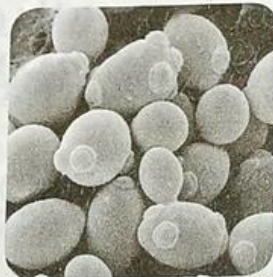
streptococcus thermophilus



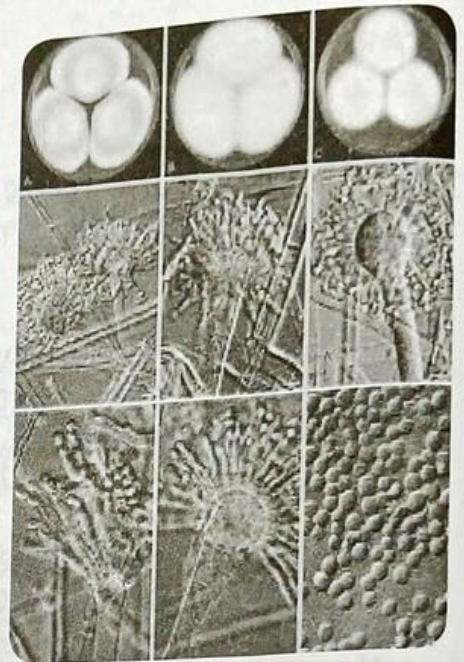
lactobacillus acidophilus



acetobacter



saccharomyces cerevisiae



aspergillus oryzae

Observation :

Sr.No	Name of the microbe	Type	Charateristics
1	Streptococcus thermophilus	Bacteria	Probiotic
2	Lactobacillus acidophilus	Bacteria	Probiotic
3	acetobacter	Bacteria	Production of vinegar
4	Saccharomyces cerevisiae	Yeast	Production of alcohol
5	Aspergillus oryzae	Fungus	Production of fermentation in soya sauce

Inference / Conclusion :

Industrial uses of microbes

- 1] Uses of microbes for garbage management
- 2] Pollution control
- 3] Production of dairy products

Multiple Choice Questions

1. Citric acid used in production of beverages, toffees, chocolates is obtained by fermentation of by *Aspergillus niger*.
a. grapes ☒ b. sugar molasses c. apple d. coffee nuts
2. The amino acid is obtained by microbial process by *Lactobacillus delbrueckii*.
a. itaconic acid ☒ b. lactic acid c. gluconic acid d. citric acid
3. Microbial enzymes aren't used in
☒ a. glass industry. b. tanning of leather. c. cheese production. d. paper industry.
4. is the antibiotic effective against tuberculosis.
a. Penicillin ☒ b. Rifamycin c. Gentamycin d. Tetracycline
5. is obtained by fermentation of apple juice.
☒ a. Cedar b. Wine c. Coffee d. Cocoa

: Exercise :

1. "Long back cheese was not considered as vegetarian food." Why?

In cheese, traditionally, an enzyme called rennet was used which is obtained from alimentary canal of cattle. Hence long back cheese was not considered as vegetarian food.

2. State the uses of xanthan gum that is obtained by microbial process.

Uses of xanthan gum:
1] Impart thickness to ice cream
2] Weedicides
3] Production of pigment

3. Why micro organisms are important in our daily life?

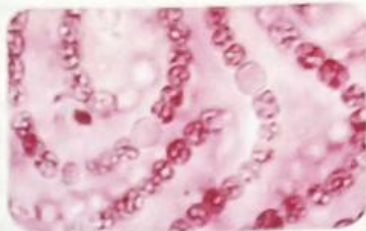
Micro organisms are very important in our daily life because they help in production of various dairy products, antibiotics, probiotics, beverages etc.

Remark and Signature



d. Nostoc

Kingdom	- Bacteria
Divison	- Cyanobacteria
Example	- Nostoc



Characteristics :

1. These bacteria look like strings embedded in gelatinous casing/covering.
2. They are found in soil, wet rocks and bottoms of water reservoirs.
3. Live independently or symbiotically. They supply nitrogen to the host when they are symbiotic.
4. Autotrophic, as the cell has chlorophyll.
5. Used as food because it contains vitamins and proteins abundantly.

Conclusion :

1. The bacteria anabaena found in azolla, mosquito fern function as biofertilizers.
2. Azotobacter living independently in soil, fix atmospheric nitrogen.
3. Nostoc - the bacteria found in soil, wet rocks and water bottoms have ability of photosynthesis.

Multiple Choice Questions

1. The wormicide character of destroys worms in water.
☒ a. azotobacter b. azolla c. nostoc d. none of these
2. The bacteria nostoc are used as food because they have abundant amount of and
☒ a. proteins and vitamin C b. carbohydrates and lipids
c. water and vitamin D d. carbohydrates and proteins
3. The bacteria supply natural manure to paddy fields.
☒ a. azolla b. anabaena c. azotobacter d. nostoc
4. Which of the following bacteria can perform photosynthesis?
☒ a. anabaena b. azotobacter c. nostoc d. non of these
5. Nitrogen fixing bacteria - anabaena - are found in the plant
☒ a. anabaena b. pistia c. salvinia d. marselia

: Exercise :

Why does the microbes plays an important role in agricultural management?

Microbes play an important role in agricultural management because they help to fix the nitrogen by the process of nitrogen fixation. This helps the plant to absorb nitrogen easily and they also helps in inoculants etc.



Q1NHQ3

Remark and Signature

Practical No. 11

Aim : To perform survey of our building regarding disaster management.

Apparatus : Note book, survey charts.

Procedure : With the help of given charts, make a survey of your home/building. Give one mark each to the proper answer. In conclusions, write total marks of each survey.

Survey Chart :

Electrical Audit Survey Chart

Sr. No.	What to be audited?	Observations Yes/ Nos / Remark	Marks
1	Is the electric connection done by licensed electrician?	Yes	1
2	Is the earthing properly done?	Yes	1
3	Is the building provided with Lightning Arrester?	Yes	1
4	Are the electrical equipments I.S.I. marked?	Yes	1
5	Are the electrical wire joints open?	No	1
6	Do the wires have proper insulations?	Yes	1
7	Is there regular checking of cables / wires around the building?	Yes	1
8	Is there are any protective grid around electric pole?	Yes	1
9	Are buckets fill with sand kept ready to extinguish electrical fire?	Yes	1
10	Do the electric equipments have three pin sockets ?	Yes	1
Total Marks			9

Construction Audit Survey Chart

Sr. No.	What to be audited?	Observations Yes/ Nos / Remark	Marks
1	Are the inner and outer surfaces of the building properly plastered and painted?	Yes	1
2	Is the construction done under the supervision of proper architect?	Yes	1
3	Are the columns/beams of the building cracked?	No	1
4	Are there any changes made to original structure at the time of renovation?	Yes	1
5	Is there any leakage through water and sewage pipes ?	No	1
6	Do the children play outdoor games on the terrace? (It causes damage and leakage to the slab)	No	1

7	Is there regular repairing and painting of the building?	Yes	1
8	Is structural audit of the building done regular?	Yes	1
9	In monsoon, does the water logs in the building area for more than 48 hours?	No	1
10	Do your area experience earthquakes regularly?	No	1
Total Marks			10

Fire Control Management Audit Survey Chart

Sr. No.	What to be audited?	Observations Yes/ Nos / Remark	Marks
1	Does the building have fire-extinguisher?	Yes	1
2	Is it in working condition?	Yes	1
3	Does staircase have spaces more than 5 square feet?	Yes	1
4	In there an alternative exit to you building?	No	0
5	Is the list of contact numbers of fire brigades kept at front side of the building?	Yes	1
6	Is there any scrap kept near the main electrical connections area?	No	1
7	Does the fire brigade arrange regular Mock Drills?	Yes	1
8	Are there any inflammable chemicals kept in buiding or home premises?	No	1
9	Is the gas cylinder kept away from fridge?	Yes	1
10	Is each floor provided with enough first aid material?	Yes	1
Total Marks			9

Conclusion :

According to above observations, your home/buiding has secured following marks

Sr. No.	Type of Audit	Marks	Remark
1	Electrical Audit	10	
2	Construction Audit	10	
3	Fire Control Management Audit	9	

Remark and Signature

Multiple Choice Questions

- At the time of fire, which of the following things we must not do?
a. Inform fire brigade
☒ c. Use lift
b. Evacuate the building as early as possible
d. Call for help from window and balcony
- of the following is manmade disaster.
a. Earthquake b. Flood c. Meteoric ☒ d. Poisonous gas leak
- We don't get information of in the disaster management clock?
a. outsiders or helpers come to know about exact location of people.
b. it becomes easier to help needy / weak people.
☒ c. local time.
d. social institutes or government can work with the local people at time suitable for them.
- What will you do, if the domestic gas cylinder bursts?
a. sprinkle water. b. throw soil or sand.
☒ c. cover the cylinder with wet bedsheet. d. run away.
- What will you do, if you experience earthquake inside the home?
☒ a. Take shelter below a table.
b. Come outside the home, in a open space.
c. Stay away from electric pole.
d. All of these.

: Exercise :

- Why the survey of our building/ home/ school regarding disaster management is necessary?
The reason behind survey of our building/home/school regarding disaster management is necessary
(i) To avoid future accidents
(ii) To aware people
(iii) To inform general public about precaution taken before or during any disaster.
- Explain any one disaster you faced. What are the efforts you used to come out from that disaster?
I have faced the disaster of gas leakage when I felt the smell of gas first of all I opened all the windows and doors I switched off electric supply. I called in gas emergency to check the leakage. After that the problem of leakage was sort out. In such cases awareness is very important.

Remark and Signature



Practical No. 12

Aim : To show that carbondioxide is evolved in the process of respiration.

Apparatus : 25-30 seeds of lentils like mung, mutki, water, cotton, conical flask, bent tube, rubber cork, etc.

Chemicals : Potassium hydroxide pallets.

Procedure :

1. Soak the given seeds in water overnight. Then keep them in wet cotton for some hours, so that they will sprout.
2. Take these wet sprout in a conical flask; so that their respiration will be continued.
3. Insert a test tube containing KOH pallets, in the conical flask.
4. Tightly fit a rubber cork with bent tube and make the flask air tight.
5. Insert the other end of bent tube in a beaker containing coloured water. Mark the level of coloured water in the bent tube.
6. Keep this set steady for 15 minutes and again marks the water level in the bent tube.
7. Note your observations.

Observation : Carbon dioxide evolved in respiration is absorbed by **KOH**, therefore water level in bent tube rises.

Inference / Conclusion : **CO₂** gas is evolved during respiration in plant.

The rise in the level of water indicates that the CO₂ is produced by the germinating seeds during respiration. The germinating seed respires and produce CO₂ which is absorbed by KOH. The air present in the glass tube moves into the conical flask this pulls the water in bent tube.

Multiple Choice Questions

1. For experiment showing that CO₂ is evolved during respiration process in plants - arrange following steps in proper order.
 - i) Soak gram seeds in water, overnight. Keep them in conical flask.
 - ii) Close the conical flask with cork fitted with 'U' tube.
 - iii) Insert test tube with KOH solution in the conical flask.
 - iv) Keep the set steady for some time.

a. i,ii,iii,iv b. i,ii,iv,iii ✓ c. i,iii,ii,iv d. i,iii,iv,ii
2. Decomposition is a type of
 - a. aerobic respiration
 - b. anaerobic respiration ✓
 - c. exothermic reaction
 - d. non of these.

3. Mohan prepared lime water and next month used it for the experiment showing CO_2 is evolved in the process of respiration.
- White precipitate is formed.
 - White precipitate is not formed.
 - Lime water must be fresh.
 - Lime water can be used anytime.
- a. (ii) is correct. ☒ b. (ii) and (iii) are correct.
c. (i) and (iv) are correct. d. (ii) and (iv) are correct.
4. Before arranging the experiment "Carbon dioxide evolved during the process of respiration", the seeds must be
- a. completely dried. b. boiled to soften them.
c. soaked in vinegar. ☒ d. kept moist till germination.
5. The seeds respire at the time of germination. While showing this experiment a test tube with KOH is kept in the conical flask to
- a. supply oxygen necessary for respiration.
☒ b. create space in the conical flask by absorbing CO_2 .
c. absorb water from the seeds and dry them.
d. make the air in the conical flask, acidic.

: Exercise :

1. How do the process of respiration occur in living organisms?

Respiration is the phenomena where the cells obtain energy by breaking down glucose. This process occurs by cellular respiration which produces oxygen is utilized to break down glucose which produces high amount of energy.

2. Why do we tired on quick exercise?

During constant exercise very large amount of energy is required for this energy production oxygen also requires. During constant exercise our muscle cells perform anaerobic respiration which produces lactic acid. This lactic acid builds up in our body and we feel tired.

3. Which type of respiration involves the process of fermentation?

Anaerobic respiration involves the process of fermentation along with glycolysis. In anaerobic respiration only 2 ATP energy is produced.

Remark and Signature

